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Presented by:
Ahmed Amine Tiffrent

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Dr. Bendaoud Fayssal	President
Dr. Khaldi Belkacem	Supervisor
Dr. Bousmaha Rabab	Examiner

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Abstract

Ensuring fish freshness is an essential aspect of the seafood industry, as it directly affects food safety, consumer satisfaction, and economic value. Traditional assessment methods rely primarily on human expertise, which, despite experience, remains subjective and prone to inconsistencies. Variations in lighting conditions, storage environments, and handling methods can lead to different evaluations of freshness, making it difficult to standardize quality control procedures.

This project aims to develop an AI-powered system capable of providing an objective, reliable, and scalable evaluation of fish freshness by analyzing visual indicators. By leveraging advanced image processing techniques, the system processes images of fish and extracts key freshness attributes to deliver an automated classification. The proposed approach eliminates human bias, enhances efficiency, and ensures consistent quality evaluation, making it a valuable tool for the seafood industry.

The solution is designed to be accessible and practical, with applications in fisheries, wholesale markets, and retail environments. It allows businesses to maintain high-quality standards, reduce waste, and optimize supply chain operations by ensuring only the freshest fish reach consumers. Additionally, consumers can benefit from a transparent and scientific method of assessing seafood quality before making purchasing decisions. By integrating AI into seafood quality assessment, this project contributes to modernizing the industry and promoting better food safety practices.

Keywords: AI, Fish Freshness, Automated Quality Control, Computer Vision, Seafood Industry, Food Safety, Supply Chain Optimization.

Résumé

Garantir la fraîcheur du poisson est un enjeu essentiel dans l'industrie des produits de la mer, car elle influence directement la sécurité alimentaire, la satisfaction des consommateurs et la valeur économique. Les méthodes d'évaluation traditionnelles reposent principalement sur l'expertise humaine, qui reste subjective et sujette à des incohérences malgré l'expérience. Les variations des conditions d'éclairage, de stockage et de manipulation peuvent entraîner des évaluations différentes de la fraîcheur, rendant difficile la standardisation des procédures de contrôle qualité.

Ce projet vise à développer un système intelligent basé sur l'intelligence artificielle (IA) capable de fournir une évaluation objective, fiable et évolutive de la fraîcheur du poisson à partir d'indicateurs visuels. Grâce à des techniques avancées de traitement d'image, le système analyse les images de poisson et extrait des attributs clés pour une classification automatisée de la fraîcheur. L'approche proposée élimine le biais humain, améliore l'efficacité et garantit une évaluation cohérente de la qualité, en faisant un outil précieux pour l'industrie halieutique.

La solution est conçue pour être accessible et pratique, avec des applications dans les pêcheries, les marchés de gros et les points de vente. Elle permet aux entreprises de maintenir des normes de qualité élevées, de réduire le gaspillage et d'optimiser la chaîne d'approvisionnement en garantissant que seuls les poissons les plus frais atteignent les consommateurs. De plus, les consommateurs bénéficient d'une méthode scientifique et transparente pour évaluer la qualité des produits de la mer avant l'achat. En intégrant l'IA dans le contrôle de qualité des produits de la mer, ce projet contribue à moderniser le secteur et à promouvoir de meilleures pratiques en matière de sécurité alimentaire.

Mots-clés : Intelligence Artificielle, Fraîcheur du Poisson, Contrôle Qualité Automatisé, Vision par Ordinateur, Industrie Halieutique, Sécurité Alimentaire, Optimisation de la Chaîne Logistique.

الملخص

تُعد ضمان جودة الأسماك الطازجة أمرًا بالغ الأهمية في قطاع المنتجات البحرية، حيث تؤثر بشكل مباشر على سلامة الغذاء ورضا المستهلكين والقيمة الاقتصادية. وتعتمد الطرق التقليدية للتقييم بشكل أساسي على الخبرة البشرية، والتي بالرغم من أهميتها تظل عرضة للتحيّز والتفاوتات. إذ يمكن أن تؤدي الاختلافات في الإضاءة وظروف التخزين وطرق المعالجة إلى تباين في تقييم درجة الطزاجة، مما يُصعّب توحيد إجراءات مراقبة الجودة.

يهدف هذا المشروع إلى تطوير نظام ذكي مدعوم بالذكاء الاصطناعي قادر على تقديم تقييم موضوعي وموثوق وقابل للتوسّع لطزاجة الأسماك من خلال تحليل المؤشرات البصرية. وبالاعتماد على تقنيات معالجة الصور المتقدمة، يقوم النظام بتحليل صور الأسماك واستخلاص الخصائص الدالة على الطزاجة لتصنيفها بشكل تلقائي. وتلغي هذه المقاربة التحيز البشري، وتعزز من كفاءة التقييم، وتضمن ثباتًا في جودة الفحص، مما يجعلها أداة قيمة لقطاع الأسماك.

تم تصميم هذا النظام ليكون سهل الاستخدام وعملي، وقابل للتطبيق في المصايد، والأسواق بالجملة، ومحلات البيع بالتجزئة. كما يساعد المؤسسات على الحفاظ على معايير جودة عالية، وتقليل الهدر، وتحسين سلسلة التوريد عبر ضمان وصول الأسماك الطازجة فقط للمستهلكين. إضافةً إلى ذلك، يمكن للمستهلكين الاستفادة من وسيلة علمية وشفافة لتقييم جودة المنتجات البحرية قبل الشراء. ومن خلال دمج الذكاء الاصطناعي في تقييم جودة المنتجات البحرية، يسهم هذا المشروع في تحديث الصناعة وتعزيز ممارسات سلامة الغذاء.

الكلمات المفتاحية: الذكاء الاصطناعي، طزاجة الأسماك، مراقبة الجودة الآلية، الرؤية الحاسوبية، صناعة المنتجات البحرية، سلامة الغذاء، تحسين سلسلة التوريد.

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LIST OF ACRONYMS

AI	Artificial Intelligence.
ANN	Artificial Neural Network.
AutoML	Automated Machine Learning.
CNN	Convolutional Neural Network.
CPU	Central Processing Unit.
DL	Deep Learning.
DMA	Dimethylamine.
E-Nose	Electronic Nose.
E-Tongue	Electronic Tongue.
GPU	Graphics Processing Unit.
Grad-CAM	Gradient-weighted Class Activation Mapping.
KNN	k-Nearest Neighbors.
LDA	Linear Discriminant Analysis.
LSTM	Long Short-Term Memory.
ML	Machine Learning.
MobileNet	Lightweight Deep Learning Model.
NNs	Neural Networks.
PCA	Principal Component Analysis.
ResNet	Residual Neural Network.
RF	Random Forest.
RNN	Recurrent Neural Network.
SHAP	SHapley Additive Explanations.
SqueezeNet	Compact CNN Architecture.
SVM	Support Vector Machine.

TF Lite	TensorFlow Lite.
TMA	Trimethylamine.
TVB-N	Total Volatile Base Nitrogen.

INTRODUCTION

Motivation

Ensuring the freshness of fish is a critical factor in the seafood industry, directly impacting food safety, consumer satisfaction, and economic value. The perishable nature of fish makes rapid quality degradation inevitable, requiring reliable and accurate methods to assess freshness. Traditional methods such as sensory evaluation, which rely on human expertise, are often subjective, inconsistent, and unsuitable for large-scale operations. With advancements in Artificial Intelligence (AI), particularly in computer vision and deep learning, it has become possible to automate freshness assessment, reducing human bias and increasing scalability. This shift holds significant potential for modernizing quality control processes in fisheries and the seafood supply chain.

Problem & Solution

The primary problem in fish freshness assessment lies in the reliance on manual, subjective, and time-consuming techniques. Variations in environmental conditions, individual expertise, and storage practices contribute to inconsistent results. Moreover, traditional chemical and microbial testing methods, while accurate, are costly and impractical for real-time or consumer-level use.

The proposed solution is to leverage computer vision and deep learning technologies to develop an AI-powered system capable of analyzing visual indicators such as fish eyes, gills, and skin. This system will provide objective, consistent, and efficient freshness classifications (e.g., *Fresh*, *Moderately Fresh*, *Spoiled*). By integrating this technology into a web or mobile application, the system can be accessible to fisheries, retailers, and consumers alike.

Research Objectives

- Develop a deep learning-based system for fish freshness classification using computer vision techniques.
- Analyze key visual freshness indicators (e.g., fisheye clarity, gill color, skin texture) and their correlation with quality levels.

- Create a scalable, user-friendly web or mobile application for real-time fish freshness assessment.
- Address the limitations of traditional methods by providing an automated, non-invasive, and consistent solution.

Limitations

- The model's accuracy may vary across different fish species, requiring species-specific adjustments.
- Dependence on high-quality input images; suboptimal lighting or blurry images may reduce prediction reliability.
- Limited applicability to non-visual freshness indicators (e.g., smell or texture) not captured by computer vision.

Thesis Organization

This thesis is organized into nine chapters as follows:

- **Chapter 1: Foundations of Artificial Intelligence and Image Processing** This chapter introduces the core principles of Artificial Intelligence (AI), Machine Learning (ML), and Deep Learning (DL), with an emphasis on their relevance to image-based classification. It explains neural networks, convolutional neural networks (CNNs), and their advantages in image recognition tasks. The chapter also covers essential image processing techniques such as normalization, resizing, cropping, data augmentation, and feature extraction, forming the backbone of AI-powered visual analysis.
- **Chapter 2: AI in Quality Assessment** This chapter explores the practical applications of AI in assessing food quality, particularly in meat, fruits, vegetables, and fish. It discusses key visual and chemical indicators of freshness and how AI models—such as computer vision and sensor-based systems—enable accurate and non-invasive assessment. Challenges in AI-based classification such as class imbalance, generalization across fish species, and lighting variations are also addressed, along with recent advancements like transfer learning and explainability tools.
- **Chapter 3: Methods for Fish Freshness Detection** This chapter categorizes the main AI-powered methods used for detecting fish freshness: computer vision techniques, Electronic Nose (E-Nose) systems, and Electronic Tongue (E-Tongue) systems. It explains how each method works, provides application examples, and discusses their advantages and limitations in real-world scenarios.
- **Chapter 4: Comparison of Methods** This chapter delivers both a general and specific comparison of the fish freshness detection methods covered in Chapter 4. It evaluates the principles behind each method, the AI technologies used, and their strengths and weaknesses. A dedicated section compares deep learning-based computer vision models in terms of classification accuracy, explainability (via Grad-CAM), and deployment challenges, offering a clear perspective on which methods are most practical and effective.

- **Chapter 5: Conclusion** The final chapter summarizes the research findings, highlighting the most effective AI-based techniques for fish freshness detection. It reflects on the limitations encountered during the project and outlines potential directions for future work, such as integrating multimodal data sources and expanding the dataset for improved generalization.

Part I

Background

CHAPTER 1

FOUNDATIONS OF ARTIFICIAL INTELLIGENCE AND IMAGE PROCESSING

1.1 Introduction

Artificial Intelligence (AI) has become a transformative force across various fields, enabling machines to mimic human cognitive functions such as perception, learning, and decision-making. Within AI, Machine Learning (ML) and Deep Learning (DL) have emerged as powerful subfields, allowing systems to learn patterns from data and make intelligent predictions.

In recent years, these technologies have played a significant role in computer vision and image processing, enabling applications ranging from medical diagnostics to autonomous vehicles. Deep learning models, especially Convolutional Neural Networks (CNNs), have demonstrated exceptional capabilities in image classification, object detection, and segmentation.

This chapter presents the foundational concepts of AI and its subfields, followed by key techniques in image preprocessing and feature extraction. It also introduces evaluation metrics essential for measuring the performance of AI models in classification tasks. The goal is to establish a strong theoretical foundation that supports the development of intelligent vision-based systems, such as the fish freshness classification system explored in this study.

Throughout this chapter, practical techniques are introduced alongside theoretical concepts, providing a comprehensive understanding of how AI and image processing intersect to solve real-world problems effectively.

1.2 Foundations of Artificial Intelligence

Artificial Intelligence (AI) is a multidisciplinary field at the intersection of computer science, mathematics, neuroscience, and cognitive science. It aims to design systems that can mimic human cognitive functions, including reasoning, learning, perception, and decision-making. Within the domain of AI, two major subfields have emerged and revolutionized the development of intelligent systems: Machine Learning (ML) and Deep Learning (DL).

1.2.1 Defining AI, ML, and DL

1.2.1.1 Artificial Intelligence (AI):

AI is the broadest term referring to machines or systems that can simulate aspects of human intelligence. The capabilities of AI range from rule-based expert systems to sophisticated models that can interpret images, play chess, or generate human-like language [1].

1.2.1.2 Machine Learning (ML):

ML is a subset of AI that focuses on algorithms that improve automatically through experience. Instead of being explicitly programmed, ML models learn patterns from data to make predictions or decisions. The key categories include:

- **Supervised Learning:** Models are trained using labeled datasets.
- **Unsupervised Learning:** Models discover hidden patterns in data without labels.
- **Reinforcement Learning:** Agents learn by interacting with an environment and receiving feedback.

1.2.1.3 Deep Learning (DL):

DL is a subfield of ML based on artificial neural networks with many layers (deep architectures). It is particularly effective for unstructured data such as images, audio, and text. DL models can automatically extract hierarchical features and are the foundation of modern AI breakthroughs, such as image classification, speech recognition, and generative models [2].

1.2.2 Neural Networks and Learning Mechanisms

1.2.2.1 Artificial Neural Networks (ANNs):

ANNs are computational models inspired by the human brain's network of neurons. They consist of layers: input, hidden, and output. Each neuron receives input, applies a transformation (usually a weighted sum followed by an activation function), and passes the result to the next layer. ANNs are the building blocks of many ML and DL models [2].

1.2.2.2 Training and Optimization:

Neural networks are trained using datasets with the goal of minimizing prediction error. This process begins with forward propagation, where the network computes output predictions based on the input data. The loss function is then used to measure the difference between the predicted outputs and the true values. To improve the model, backpropagation calculates the gradients of the loss with respect to the network's weights. Finally, gradient descent is applied to update the weights in a direction that minimizes the loss, progressively enhancing the network's performance.

1.2.3 Convolutional Neural Networks (CNNs)

CNNs are specialized neural networks designed for processing visual data. Unlike traditional ANNs, CNNs leverage the spatial structure of images. They use convolutional layers to extract features such as edges and shapes, pooling layers to reduce dimensionality, and fully connected layers for classification. The overall architecture of a typical CNN used for image classification is illustrated in Figure 1.1.

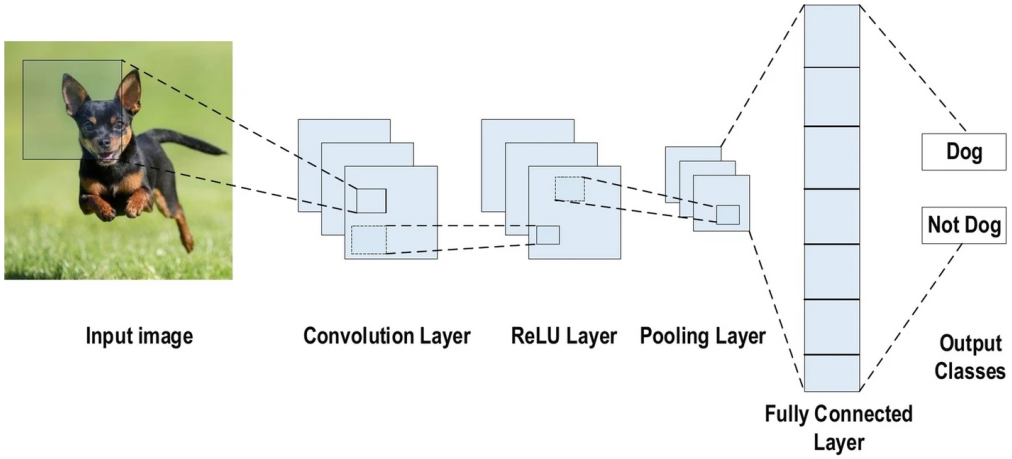


Figure 1.1: CNN architecture for image classification [3]

1.2.3.1 Advantages of CNNs:

- Automatic feature extraction from raw image data.
- Parameter sharing and local connectivity, reducing complexity.
- Translation and scale invariance.

CNNs are especially effective in image-based tasks, such as those used in this project for classifying fish freshness based on eye characteristics.

1.2.4 Real-World Applications of AI

Artificial Intelligence (AI), Machine Learning (ML), and Deep Learning (DL) have penetrated nearly every domain of modern technology. In healthcare, they are used for diagnosis through medical imaging, drug discovery, and personalized treatment plans. In the finance sector, they enhance risk assessment, detect fraud, and enable algorithmic trading. Transportation has seen advancements through the development of autonomous vehicles and improved traffic management systems.

In retail, intelligent systems analyze customer behavior and power recommendation engines. Furthermore, in environmental science, they are applied to predict climate changes and monitor wildlife populations. These wide-ranging applications highlight the flexibility and profound impact of intelligent systems across industries.

1.2.5 AI in Fish Freshness Estimation

In this project, we leverage AI—specifically CNN-based deep learning—for the automated assessment of fish freshness. This approach enables accurate, fast, and objective grading, replacing subjective and time-consuming traditional methods. The system processes images of fish eyes and classifies them into freshness levels: Fresh, Moderately Fresh, or Spoiled.

> The fusion of AI and computer vision enables transformative applications in food quality control, and this project exemplifies its use in the fisheries industry.

1.3 Image Processing Techniques in Computer Vision

Image processing is a fundamental step in the computer vision pipeline, as it prepares and enhances images to improve the performance of machine learning and deep learning models. This section explores key techniques used in image preprocessing and feature extraction, which play a crucial role in improving model accuracy and generalization.

1.3.1 Image Preprocessing Techniques

Preprocessing techniques ensure that input data is consistent, noise-free, and tailored to the input requirements of neural networks. Proper preprocessing helps reduce computational costs, improve model convergence, and standardize data across the dataset.

1.3.1.1 Normalization

Normalization scales pixel values to a specific range (commonly [0, 1]) to reduce inconsistencies in image intensities caused by lighting variations. Deep learning models such as CNNs benefit from normalized inputs as it accelerates convergence and avoids gradient-related issues.

Common Normalization Methods:

- **Min-Max Normalization:**
$$x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}}$$
- **Z-score Normalization:**
$$x' = \frac{x - \mu}{\sigma}$$

1.3.1.2 Image Resizing and Cropping

Neural networks require fixed-size inputs. Resizing ensures all input images are of the same shape (e.g., 224×224 or 299×299), while cropping helps remove irrelevant background, focusing on the object of interest.

Two cropping strategies were employed to optimize the model's ability to focus on relevant visual features. Center cropping was used to systematically remove border regions and retain the central portion of the image, where the fish eye is typically located. This ensures that critical anatomical details are preserved. On the other hand, random cropping was applied during training to introduce variability and improve the model's robustness. By exposing the model to slightly different portions of the image in each epoch, random cropping helps mitigate overfitting and enhances generalization.

1.3.1.3 Data Augmentation

Data augmentation increases the diversity of training data by applying random transformations. This simulates real-world variations and improves the generalization ability of the model.

To improve the model's generalization and robustness, several common image augmentation techniques were applied during training. Rotation was used to simulate different orientations of the fish eye, ensuring the model learns to recognize features from multiple angles. Flipping, both horizontal and vertical, introduced mirrored perspectives to further increase data diversity. Brightness and contrast adjustments mimicked varying lighting conditions typically encountered in real-world settings. Lastly, zooming and shearing operations helped the model adapt to scale changes and subtle perspective shifts, making it more resilient to variability in image capture.

1.3.2 Feature Extraction Techniques

After preprocessing, feature extraction identifies the most informative aspects of an image. These features serve as inputs for classification models, helping to reduce dimensionality and focus on essential patterns.

1.3.2.1 Edge Detection

Edge detection algorithms highlight transitions in intensity, which often correspond to object boundaries or texture changes.

Several classical edge detection algorithms were explored to highlight structural features within fish eye images. The Sobel operator calculates intensity gradients in horizontal and vertical directions, effectively capturing edge transitions. The Canny edge detector employs a multi-stage process—including noise reduction, gradient calculation, non-maximum suppression, and edge tracking—to produce accurate and robust edge maps. Additionally, the Prewitt operator, which is conceptually similar to Sobel, offers a simpler approach for detecting gradients and delineating basic edge structures in images.

1.3.2.2 Color Analysis

Color analysis helps in classifying and segmenting images based on hue, saturation, and brightness, which can indicate freshness or ripeness in biological images.

Color-based feature extraction techniques are commonly used in image analysis. Color histograms provide a quantitative representation of the distribution of colors within an image, which can be useful for distinguishing freshness levels based on visual cues. Additionally, HSV color segmentation allows for more effective color isolation, enabling the model to detect specific hues such as redness, which is often associated with spoilage in fish.

1.3.2.3 Texture and Shape Features

For tasks such as fish freshness estimation, analyzing the texture of gills or the shape of the eyes can be crucial.

Texture-based feature extraction methods are essential for capturing surface patterns and fine details in images. Gabor filters are widely used to extract texture information by analyzing image content at various orientations and scales, making them suitable for biological textures such as those found in fish eyes. Meanwhile, the Histogram of Oriented

Gradients (HOG) technique encodes the structure and shape of objects by computing the distribution of gradient directions, which helps in identifying visual patterns relevant to classification tasks.

1.3.3 Role of Image Processing in Fish Freshness Estimation

In our system, image processing ensures that fisheye images are clean, standardized, and informative. Techniques such as resizing, normalization, and augmentation improve model learning. Feature extraction from eye clarity, color intensity, and shape contributes to the accurate classification of freshness into categories like *Fresh*, *Moderately Fresh*, or *Spoiled*. The examples are illustrated in Figure 1.2.

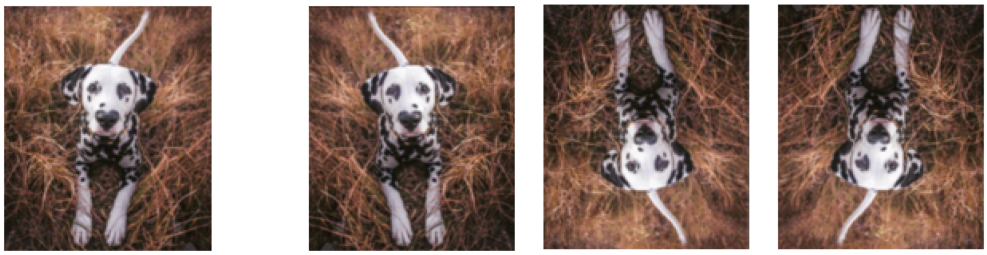


Figure 1.2: Examples of data augmentation techniques [4]

1.4 Evaluation Metrics for AI Models

Evaluating the performance of AI models is essential to understanding how well they perform on real-world data. In classification tasks, such as fish freshness prediction, different metrics provide different insights. This section explains the most commonly used evaluation metrics, their mathematical definitions, and their relevance in practice.

1.4.1 Classification Metrics Overview

When evaluating models in supervised classification tasks, we use several metrics to assess different aspects of performance. These metrics rely on the confusion matrix, which summarizes the counts of:

- **True Positives (TP):** Correctly predicted positive cases.
- **True Negatives (TN):** Correctly predicted negative cases.
- **False Positives (FP):** Incorrectly predicted positive cases.
- **False Negatives (FN):** Incorrectly predicted negative cases.

1.4.1.1 Accuracy

Accuracy measures the proportion of correctly predicted samples out of the total number of predictions:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

It provides a general idea of model performance but may be misleading in imbalanced datasets.

1.4.1.2 Precision

Precision focuses on the quality of positive predictions:

$$\text{Precision} = \frac{TP}{TP + FP}$$

It answers the question: "Of all predicted positives, how many were correct?"

1.4.1.3 Recall (Sensitivity)

Recall, also known as sensitivity or true positive rate, measures the model's ability to detect all positive samples:

$$\text{Recall} = \frac{TP}{TP + FN}$$

It answers the question: "Of all actual positives, how many did the model correctly identify?"

1.4.1.4 F1 Score

The F1 score is the harmonic mean of precision and recall. It provides a balanced measure when there is a trade-off between the two:

$$\text{F1 Score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

F1 is especially useful in imbalanced classification problems, where accuracy alone can be deceptive.

1.4.1.5 Confusion Matrix

The confusion matrix is a visual tool used to summarize prediction results. It shows the counts of actual vs predicted labels, making it easier to see where misclassifications occur. Each row represents the actual class, while each column represents the predicted class.

1.4.1.6 Macro vs Micro Averaging (Multiclass)

For multiclass classification (e.g., Fresh, Moderately Fresh, Spoiled), we compute macro-averaged and micro-averaged metrics:

- **Macro-average:** Calculates metric independently for each class, then takes the average. It treats all classes equally.
- **Micro-average:** Aggregates contributions of all classes to compute the metric globally. It favors larger classes.

1.4.2 ROC Curve and AUC (for Binary Classification)

The Receiver Operating Characteristic (ROC) curve plots the true positive rate (TPR) against the false positive rate (FPR) at different classification thresholds. The Area Under the Curve (AUC) summarizes the ROC curve into a single score:

$$\text{AUC} \in [0.5, 1.0]$$

A model with AUC = 1.0 is perfect; AUC = 0.5 is random guessing. ROC-AUC is primarily used for binary classification.

1.4.3 Why These Metrics Matter in Our Project

In our fish freshness classification system, we rely on accuracy, precision, recall, and the F1 score to measure how well the model performs in distinguishing between freshness levels. A confusion matrix and F1 score help detect which classes the model struggles with most — such as misclassifying moderately fresh as spoiled, or vice versa — which is critical for real-world application.

Example: If the model has high accuracy but low recall for the "Spoiled" class, it may frequently fail to identify spoiled fish — which is unacceptable in food safety scenarios. Thus, a combination of metrics provides a holistic view of model performance.

CHAPTER 2

AI IN QUALITY ASSESSMENT

2.1 Applications of AI in Food Quality Control

Artificial Intelligence (AI) has become a game-changer in food quality control by enabling faster, more accurate, and scalable solutions for assessing the freshness and safety of food products. Traditionally, food inspection methods relied heavily on human sensory evaluation or laboratory testing, which can be subjective, labor-intensive, and impractical for large-scale operations. The integration of AI—particularly machine learning (ML) and deep learning (DL)—has opened new avenues for automating quality assessment through image analysis, sensor data interpretation, and pattern recognition [2, 5].

Modern AI systems can detect subtle variations in texture, color, and chemical composition that are invisible to the human eye. These systems not only improve the accuracy of assessments but also reduce operational costs and ensure consistency in food safety standards.

This section explores AI applications in the evaluation of meat, fruits, vegetables, and fish—highlighting the methods, technologies, and benefits associated with each domain. Figure 2.1 presents a general overview.

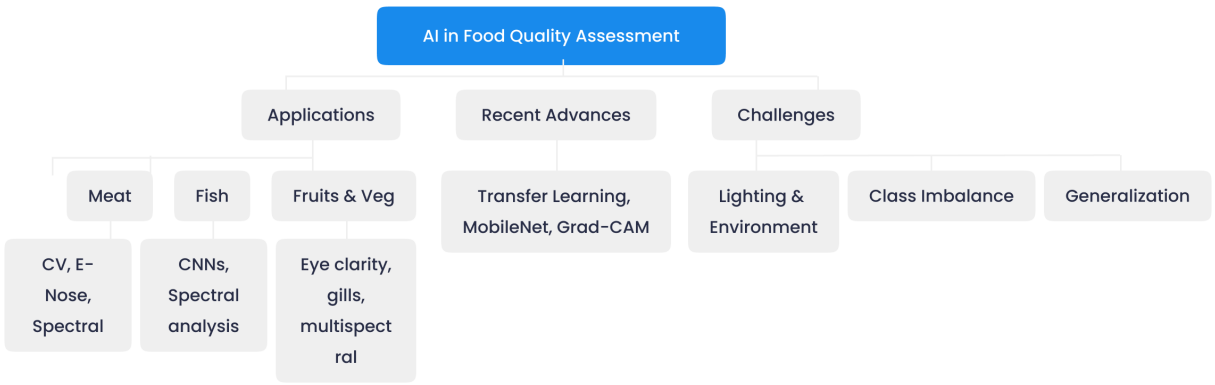


Figure 2.1: AI in quality assessment

2.1.1 Meat Freshness Detection

Meat freshness is vital for ensuring food safety, nutritional value, and consumer trust. Spoiled meat can harbor harmful bacteria such as Salmonella or E. coli, posing serious health risks. AI-based techniques now offer objective, non-invasive, and real-time meat evaluation using visual and chemical cues [6].

2.1.1.1 Techniques Used in AI-Based Meat Freshness Detection:

- **Computer Vision:** Deep CNNs analyze images of meat for changes in surface color, texture granularity, and marbling patterns to estimate spoilage levels.
- **Spectral Imaging:** AI interprets hyperspectral data to detect internal changes in water content and myoglobin concentration—indicators of freshness.
- **Electronic Nose (E-Nose):** AI-driven olfactory sensors mimic human smell to detect volatile compounds produced during decomposition.

2.1.1.2 Benefits:

Computer vision and AI techniques provide rapid and accurate meat classification without the need for destructive sampling. They enable continuous monitoring throughout slaughterhouses and cold chain logistics, ensuring that freshness and quality standards are maintained. Additionally, these technologies help businesses ensure regulatory compliance and reduce the risk of costly product recalls.

2.1.2 Fruit and Vegetable Quality Estimation

The appearance and internal quality of fruits and vegetables are key determinants of market value and consumer satisfaction. AI plays a central role in automating produce grading, sorting, and defect detection by leveraging visual and spectral cues [7].

2.1.2.1 Key AI Techniques for Fruit and Vegetable Quality Estimation:

Image Processing techniques are employed wherein machine learning models extract color histograms and texture descriptors to assess ripeness levels or detect bruises and other defects in produce. In parallel, **Convolutional Neural Networks (CNNs)** are widely used for classification tasks, enabling the identification of overripe, spoiled, or damaged produce through detailed surface anomaly detection. Additionally, **Spectroscopy and Sensor Fusion** approaches integrate data from sources such as near-infrared (NIR) spectroscopy, moisture sensors, and traditional visual inputs, allowing AI models to deliver a more holistic and accurate prediction of overall produce quality.

2.1.2.2 Advantages:

Computer vision systems play a crucial role in the agricultural sector by reducing post-harvest losses through the early identification of defective items. They enhance grading accuracy in packing and distribution centers, ensuring that only high-quality produce reaches consumers. Furthermore, these technologies facilitate dynamic pricing strategies and improve shelf-life prediction, optimizing both inventory management and profitability.

2.1.3 Adaptation to Fish Freshness Detection

Fish freshness degrades rapidly after harvesting, making it essential to evaluate its quality promptly and accurately. Traditional methods like the Torry sensory scale or chemical assays (e.g., TVB-N, TMA) are reliable but slow and labor-intensive. AI models, particularly CNN-based vision systems, offer faster, scalable alternatives suitable for use in processing facilities, markets, and even consumer mobile apps [8].

2.1.3.1 Visual and Chemical Indicators of Fish Freshness:

Eye Clarity and Convexity serve as critical indicators of fish freshness, where a fresh fish typically displays clear, convex eyes, while cloudiness and a flattened appearance suggest the onset of spoilage. Similarly, **Gill Color and Texture** are vital parameters; bright red gills are associated with high freshness, whereas darkened, slimy, or discolored gills point to progressive decay. Lastly, **Skin Glossiness and Firmness** provide visual and tactile cues, with the loss of surface shine and a decrease in skin firmness signaling a decline in overall fish quality.

2.1.3.2 AI Approaches for Fish Freshness Estimation:

Computer Vision with CNNs is widely applied to analyze critical regions such as the fish eye and gills, classifying freshness levels with the help of pretrained convolutional models like ResNet or EfficientNet. This approach enables automated, objective evaluations based on subtle visual features. In parallel, **Electronic Nose (E-Nose)** technologies detect gases related to spoilage, such as hydrogen sulfide and ammonia, using AI algorithms to interpret data from gas sensor arrays with high precision. Furthermore, **Multispectral Imaging** merges reflectance and fluorescence information, enhancing the detection of early biochemical spoilage markers that are not visible to the naked eye. By integrating AI models, multispectral imaging offers a deeper and more sensitive analysis of food freshness.

2.1.3.3 Benefits:

Computer vision-based freshness assessment offers an objective and repeatable method for evaluating food quality. It enables non-invasive quality checks during transportation and sale, ensuring that products maintain their integrity throughout the supply chain. Additionally, this approach aligns with sustainable practices by minimizing food waste and reducing the need for chemical preservatives.

In conclusion, AI is revolutionizing food quality control by offering intelligent, rapid, and scalable solutions. In the case of fish, it allows real-time classification of freshness based on physical and chemical indicators, ultimately supporting both industry and consumer needs.

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2.2 Challenges in AI-Based Fish Freshness Detection

Despite the advancements in **AI** for fish freshness estimation, several challenges hinder its widespread adoption and effectiveness. These challenges primarily arise from environmental variability, data imbalances, and generalization issues across different fish species. Addressing these challenges is crucial for improving the accuracy and reliability of **AI**-based freshness detection systems [8, 2].

—

2.2.1 Variations in Lighting and Environmental Factors

Lighting conditions significantly affect the performance of **AI**-based fish freshness detection systems. Inconsistent lighting introduces variations in image brightness, contrast, and color, which can mislead machine learning models [7]. Additionally, environmental factors such as temperature and humidity influence fish appearance and can affect the accuracy of predictions.

2.2.1.1 Key Issues:

- Shadows, glare, and reflections in images distort key freshness indicators.
- Temperature fluctuations affect fish eye clarity and gill coloration.
- Inconsistent backgrounds introduce noise, leading to potential misclassification.

2.2.1.2 Potential Solutions:

- Use **adaptive image preprocessing** techniques such as histogram equalization to normalize brightness and contrast.
- Implement **data augmentation** (brightness adjustments, white balance correction) to make models robust to lighting variations.
- Employ **multi-spectral imaging** to capture more reliable spectral data for freshness estimation.

—

2.2.2 Class Imbalance in Datasets

One of the major challenges in training **AI** models for fish freshness detection is the imbalance in dataset distribution. In many cases, datasets contain significantly more images of fresh fish compared to spoiled fish, leading to biased model predictions [9].

2.2.2.1 Impact of Class Imbalance:

Models tend to favor the majority class, such as fresh fish, while often failing to accurately predict minority classes like moderately fresh or spoiled fish. This imbalance results in poor model generalization, leading to an increased number of false negatives, which can compromise food safety by incorrectly labeling spoiled fish as safe. Furthermore, imbalanced datasets reduce training effectiveness, causing models to learn suboptimal decision boundaries that hinder overall classification performance.

2.2.2.2 Techniques to Address Class Imbalance:

Several strategies can be employed to address class imbalance. **Data Augmentation** involves the synthetic generation of minority-class images through transformations such as flipping and rotation, enriching dataset diversity. **Resampling Methods** include techniques like oversampling the minority class using **SMOTE (Synthetic Minority Oversampling Technique)** or undersampling the majority class to achieve balance. **Cost-Sensitive Learning** introduces higher misclassification penalties for underrepresented classes,

encouraging the model to focus more on difficult cases. Additionally, **Class-Balanced Loss Functions** adjust the loss computation during training to compensate for unequal class distributions, leading to fairer and more accurate learning.

2.2.3 Generalization Across Fish Species and Sizes

Fish species exhibit diverse physical characteristics, including variations in eye clarity, gill structure, and body texture. **AI** models trained on specific fish species may struggle to generalize across different species and fish sizes [8].

2.2.3.1 Key Challenges:

- Variations in **eye size, color intensity, and gill structures** lead to inconsistent model predictions.
- Fish of different sizes may require different preprocessing steps to standardize features.
- Species with naturally cloudy eyes may be incorrectly classified as **"spoiled"** due to dataset biases.

2.2.3.2 Potential Solutions:

- Train models on **diverse multi-species datasets** to improve robustness.
- Use **transfer learning** with pre-trained models to adapt to different fish species.
- Implement **feature extraction standardization** techniques to ensure model consistency across fish sizes.

Improving model generalization is essential to ensure reliable **AI**-powered freshness detection across different fish species and environmental conditions.

—

2.3 Recent Advancements in AI-Based Classification

Advancements in **AI**-based classification have significantly improved the accuracy, efficiency, and interpretability of machine learning models in various applications, including fish freshness detection. These advancements include the use of **pretrained models** for transfer learning, **lightweight architectures** like **MobileNet** and **SqueezeNet**, and **explainability techniques** such as **Grad-CAM** [2, 10].

—

2.3.1 Pretrained Models for Transfer Learning

2.3.1.1 Concept of Transfer Learning:

Transfer learning allows **AI** models to leverage knowledge from **pre-trained networks**, reducing the need for large datasets and extensive training. A model trained on a large dataset like **ImageNet** can be fine-tuned for specific applications, such as fish freshness classification, by modifying its final layers [11]. See Figure 2.2.

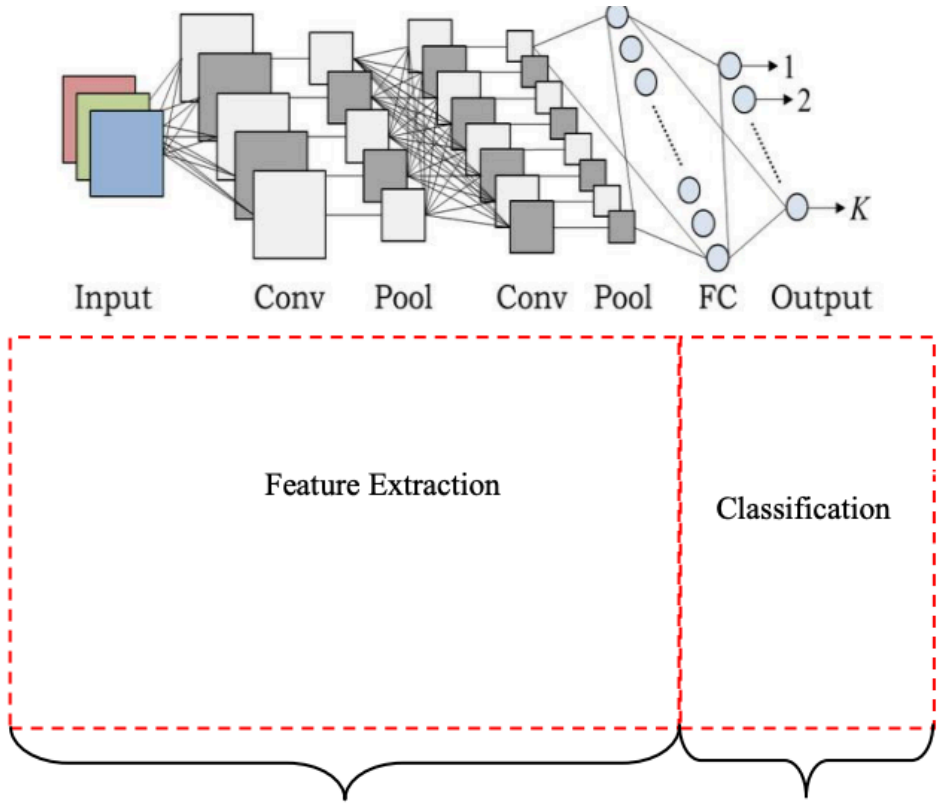


Figure 2.2: Transfer-learning [12]

2.3.1.2 Advantages:

Transfer learning offers multiple advantages in deep learning applications. It reduces computational cost and training time by leveraging pre-trained models instead of training from scratch. Additionally, it improves model accuracy on small and specialized datasets by transferring knowledge from large-scale datasets. Finally, it prevents overfitting by utilizing robust pre-learned features, ensuring that models generalize better even when limited training data is available.

2.3.1.3 Commonly Used Pretrained Models:

- **ResNet (Residual Networks)** – Suitable for deep learning tasks with minimal degradation in accuracy.
- **VGG (Visual Geometry Group Networks)** – Effective for image classification tasks.
- **EfficientNet** – Optimized for high performance with fewer parameters.

Transfer learning is particularly beneficial in fish freshness detection, where obtaining large annotated datasets is challenging.

—

2.3.2 Use of Models (e.g., MobileNet, SqueezeNet, EfficientNet...)

Lightweight models such as **MobileNet** and **SqueezeNet** have been developed to enhance computational efficiency, making **AI**-based classification feasible on edge devices and

mobile applications [13].

2.3.2.1 MobileNet:

MobileNet is designed for real-time image classification with reduced computational cost. It employs **depthwise separable convolutions** to minimize the number of parameters while maintaining high accuracy. See Figure 2.3.

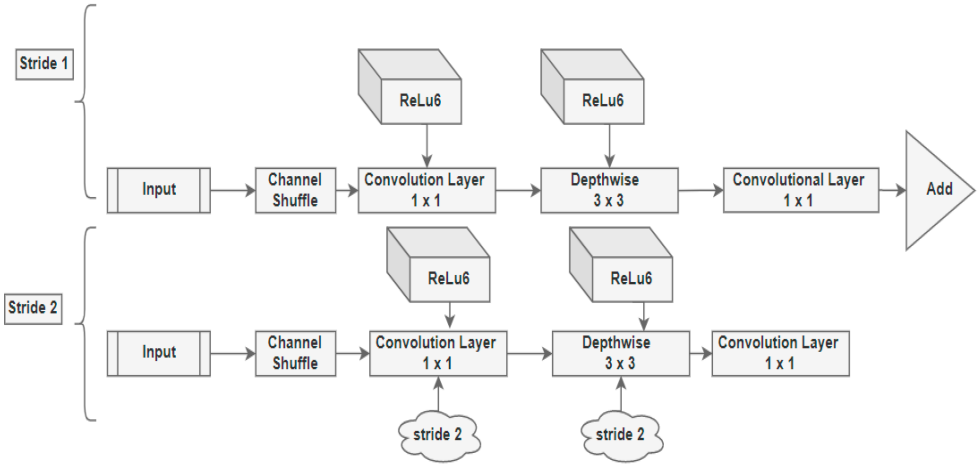


Figure 2.3: MobileNet Architecture [14]

2.3.2.2 SqueezeNet:

SqueezeNet is a compact neural network architecture that significantly reduces the number of parameters while achieving similar accuracy levels to larger models. It uses **fire modules**, which replace conventional convolutional layers with squeeze and expand operations to improve efficiency. As presented in Figure 2.4.

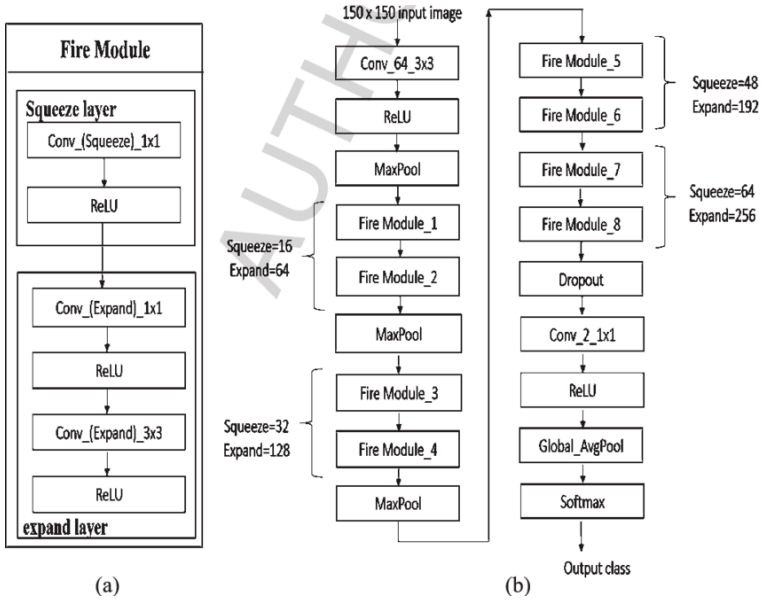


Figure 2.4: Squeeze-net Architecture [15]

2.3.2.3 EfficientNet

EfficientNet is a family of CNN architectures developed by Google that balances model accuracy and computational efficiency. It uses a compound scaling method that uniformly scales **depth, width, and resolution**. In fish freshness classification, EfficientNet has shown high performance with fewer parameters, making it ideal for applications requiring both speed and precision. As presented in Figure 2.5.

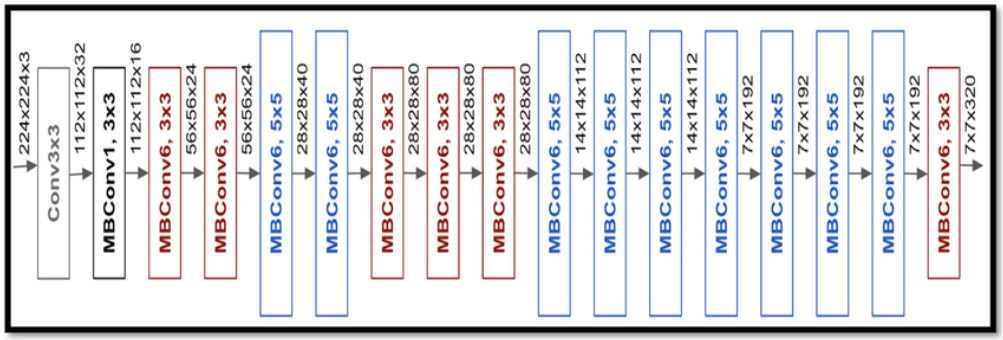


Figure 2.5: EfficientNet Architecture [16]

2.3.2.4 ResNet

ResNet (Residual Network) introduced the concept of **residual connections**, allowing much deeper networks to be trained effectively by mitigating the vanishing gradient problem. ResNet models are known for their robustness and accuracy, especially in complex classification tasks such as identifying subtle changes in fish eyes and gills. As presented in Figure 2.6.

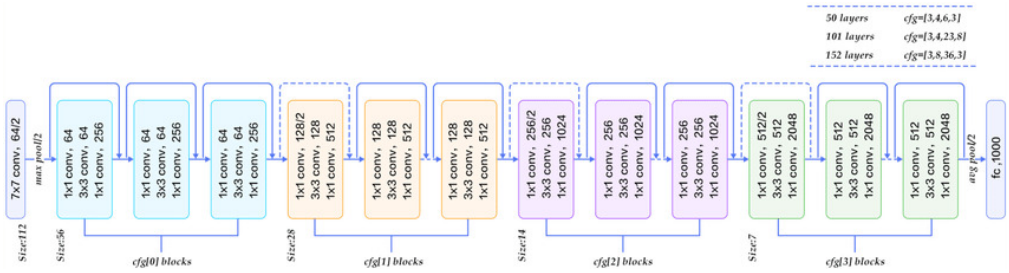


Figure 2.6: Resnet-net Architecture [17]

2.3.2.5 Advantages of Lightweight Models:

Lightweight models provide significant advantages in practical AI deployments. They offer a **reduced memory footprint**, making them suitable for mobile and embedded applications. Additionally, they deliver **faster inference time**, which is essential for real-time applications. Moreover, they have a **lower computational cost**, requiring less processing power compared to deeper architectures such as **ResNet**.

These lightweight models are ideal for fish freshness detection in real-world scenarios, where real-time analysis and deployment on portable devices are required.

—

2.3.3 Explainability Techniques (e.g., Grad-CAM)

Despite the high accuracy of deep learning models, their **black-box nature** makes it difficult to interpret how decisions are made. **Explainability techniques** such as **Gradient-weighted Class Activation Mapping (Grad-CAM)** enhance transparency in **AI**-based classification systems [18].

2.3.3.1 What is Grad-CAM?

Grad-CAM visualizes the regions of an image that influence a model's prediction. It overlays a **heatmap** on the input image, highlighting the most important features that contribute to the classification decision.

2.3.3.2 How Grad-CAM Works:

1. Computes the gradient of the output category with respect to feature maps of the last convolutional layer.
2. Weighs these gradients to understand the impact of different spatial locations.
3. Generates a heatmap that highlights **important regions** in the input image.

Figure 2.7 explains more how Grad-CAM works:

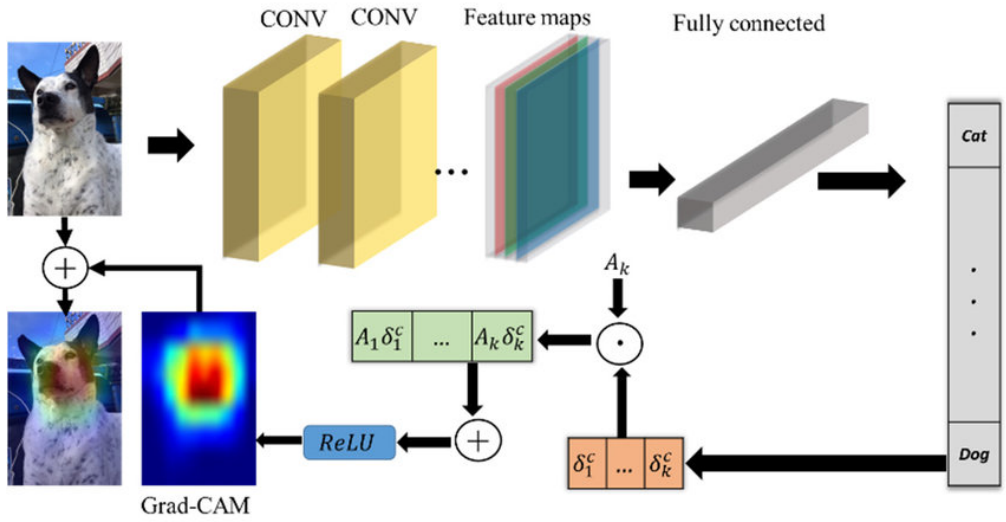


Figure 2.7: Grad-CAM [19]

2.3.3.3 Benefits of Explainability Techniques:

Visualization techniques play a critical role in enhancing the transparency of AI models. They **improve trust and interpretability** in **AI**-based classification tasks, allowing users to better understand model behavior. Additionally, they help detect **biases and model errors** during decision-making processes. Furthermore, visualizations assist researchers in refining **model architecture** and improving the effectiveness of **feature extraction**.

Grad-CAM is particularly useful in fish freshness classification, where decision-making transparency is essential for quality control and regulatory compliance.

Part II

State of Art

CHAPTER 3

METHODS FOR FISH FRESHNESS DETECTION

Fish is a highly perishable food product whose freshness directly influences food safety, nutritional value, and consumer acceptability. Traditional freshness assessment methods—such as sensory evaluation, chemical assays, and microbial analysis—are often subjective, destructive, and time-consuming. To address these limitations, recent advancements in Artificial Intelligence (AI) have introduced innovative techniques capable of performing rapid, non-invasive, and automated fish freshness classification.

This chapter provides a comprehensive overview of the three primary AI-powered methods used for fish freshness detection: Computer Vision, Electronic Nose (E-Nose), and Electronic Tongue (E-Tongue). Each technique has a unique operational principle, data modality, and application domain:

Computer Vision systems rely on image-based features such as eye clarity, gill color, and surface glossiness to assess spoilage.

E-Nose technologies simulate human olfaction by analyzing the volatile compounds released during decomposition.

E-Tongue systems mimic taste perception to detect chemical changes, such as shifts in pH or ionic composition, related to spoilage progression.

These methods are often enhanced with machine learning and deep learning algorithms that improve accuracy and support real-time classification. The Figure 3.1. below summarizes the methodological structure discussed in this chapter.

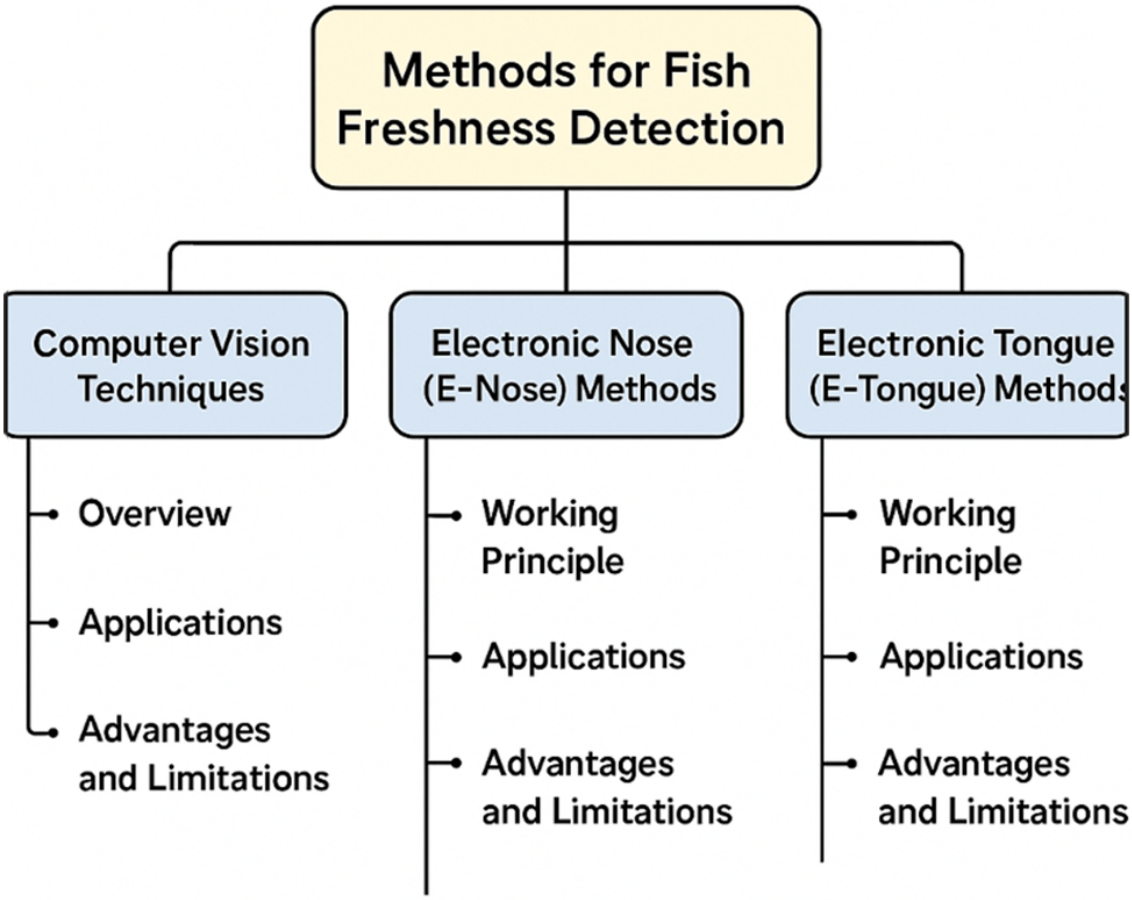


Figure 3.1: Methods for Fish Freshness Detection

3.1 Computer Vision Techniques

Fish freshness detection is a crucial aspect of food quality control. Among the various AI-based techniques, **computer vision** has emerged as a non-destructive, fast, and scalable approach for assessing fish freshness. By leveraging image processing and deep learning, computer vision systems can automate fish quality inspection with high accuracy, reducing the reliance on subjective human assessment.

3.1.1 Overview of Computer Vision in Food Quality

Computer vision is an artificial intelligence (AI) technology that mimics human vision to analyze images and extract meaningful insights. In the food industry, it is widely used for detecting **freshness, texture, color changes, and contamination** in food products. A

key advantage of computer vision is that it enables automated, real-time, and non-invasive inspection of food items.

Recent advancements have integrated deep learning models such as **Convolutional Neural Networks (CNNs)** to enhance feature extraction and classification accuracy [2]. **Computer vision in food quality control** relies on a combination of image processing techniques, including **feature extraction, segmentation, and classification**. This ensures that key quality parameters such as **color degradation, texture changes, and surface defects** can be effectively analyzed and classified.

3.1.2 Applications in Fish Freshness Detection

In fish freshness assessment, computer vision is primarily used to analyze **fisheye, gill color, body texture, and surface glossiness**, which change over time due to enzymatic degradation and microbial activity. Several studies have demonstrated the effectiveness of this approach, leveraging **machine learning** and **deep learning techniques**.

The study by **Yildiz** [20] developed a **deep learning-based fish freshness detection system** that combines the strengths of convolutional and fully connected neural networks. Specifically, the authors employed the **VGG19** architecture for **feature extraction** and used an **Artificial Neural Network (ANN)** for the classification task. The model was trained on the **Freshness of Fish Eyes (FFE) dataset**, which contains images representing different levels of freshness based on the appearance of the fish eye.

The system achieved a **77.3% classification accuracy**, demonstrating the potential of **transfer learning** in food quality assessment applications. It successfully identified visual features such as **eye cloudiness** and **shape deformation**, which are strong indicators of spoilage. The authors further suggested that integrating **multispectral imaging** techniques could enhance performance by capturing underlying **biochemical changes**, thus offering a more robust and objective evaluation of fish freshness. Figure 3.2 below present an overview of this study.

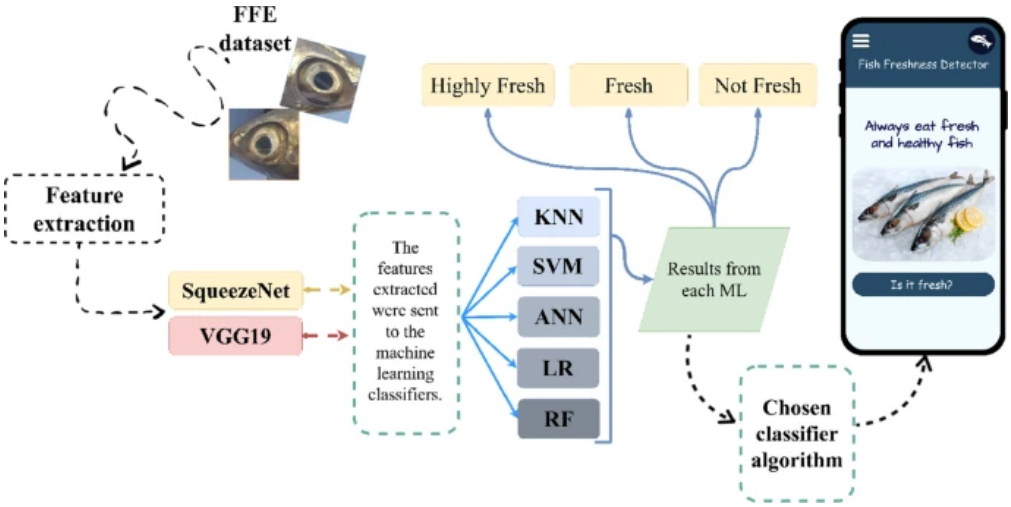


Figure 3.2: Study by Yildiz [20]

Also, a study by **Banwari** [21] proposed a **computer vision-based method** for estimating fish freshness by analyzing **segmented images of fish eyes**. This work employed classical **image processing algorithms** to extract **visual features** that correlate with

various stages of spoilage. In particular, the technique emphasized the extraction of **color and texture variations** in the eye region, which are known to deteriorate as the fish loses freshness.

The authors reported that their model achieved a **high classification accuracy**, effectively distinguishing between fresh and degraded samples. One of the major contributions of this study is its focus on **non-destructive evaluation**, offering a **fast and objective** alternative to traditional chemical tests. By eliminating the need for invasive sampling, the proposed approach is especially well-suited for **real-time quality control** applications in fish markets and processing plants. Figure 3.3 presents this study.

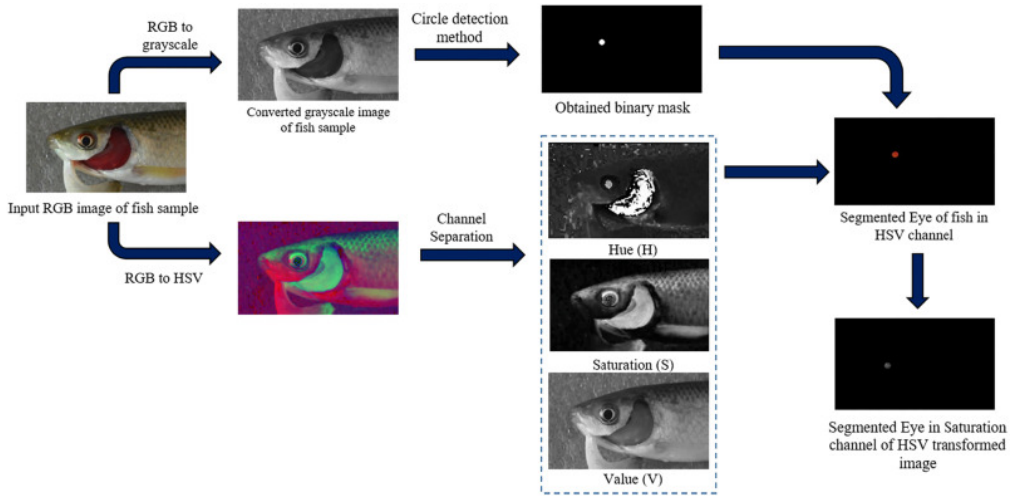


Figure 3.3: Study by Banwari [21]

The study demonstrates the potential of **targeted image analysis** for **automated fish freshness detection** and suggests future improvements using **deep learning-based segmentation techniques**.

Also, a study by **Prasetyo** [8] proposed a lightweight deep learning approach for classifying fish freshness based on eye appearance using a modified MobileNet architecture. Their model, named MobileNetV1 with Bottleneck and Expansion (MB-BE), introduces depthwise separable convolution bottlenecks combined with residual transitions to improve feature extraction efficiency. The authors conducted extensive ablation studies using the same FFE dataset and reported that their best configuration achieved a **test accuracy of 63.21%**. This result outperformed classical MobileNet, VGG16, DenseNet, and NASNet models in both performance and parameter efficiency. Although their model was less accurate than ResNet50, it required significantly fewer parameters, making it more suitable for real-time or mobile applications. This work highlights the trade-off between model size and performance and demonstrates the potential of architectural innovations in handling subtle visual differences in fish freshness classification tasks.

Alsom, a study by **Shi** [22] explored the use of **computer vision techniques** to detect fish freshness by analyzing both **chemical and biological indicators**. Their research emphasized the importance of integrating chemical metrics such as **Total Volatile Base Nitrogen (TVB-N)**, which serves as a critical marker of spoilage, and **Thiobarbituric Acid (TBA)** levels, which are associated with lipid oxidation and rancidity.

To model the relationship between visual features and chemical indicators, the authors employed **deep learning regression models**, achieving a **high correlation coefficient of 0.98**. This result demonstrated the model's strong predictive capability for assessing fish freshness. The study concluded that a hybrid approach combining **chemical analysis with image-based features** provides a more accurate and reliable system for freshness evaluation.

3.1.3 Advantages and Limitations of Computer Vision

Advantages: Computer vision-based methods offer several notable advantages for fish freshness detection. They are inherently **non-destructive and hygienic**, as they do not require direct physical interaction with the fish, unlike sensory or chemical testing. These methods are also ideal for **scalability and automation**, making them suitable for large-scale fish processing industries where automation is critical. Moreover, they deliver **high classification accuracy**, with deep learning models like **VGG19, MobileNet, and SqueezeNet** providing precise and reliable results. Another major advantage is **real-time processing**, allowing fish freshness to be classified within seconds. Finally, computer vision approaches are **cost-efficient**, eliminating the need for expensive biochemical tests and making them accessible even to small and medium-sized fish vendors.

Limitations: Despite their advantages, computer vision-based methods for fish freshness detection also face several limitations. **Lighting sensitivity** is a major issue, as variations in lighting conditions can significantly impact the accuracy of color-based freshness assessments. Additionally, these models exhibit strong **dataset dependency**, requiring large, high-quality labeled datasets to achieve optimal performance. **Generalization challenges** are also prevalent, with models trained on a specific fish species often struggling to perform well on different species. Furthermore, **computational requirements** can be prohibitive, as deep learning models demand substantial processing power, making deployment difficult in low-resource environments. Lastly, **environmental factors** such as temperature, humidity, and camera angle variations can introduce noise into image analysis, further affecting prediction reliability.

Despite these challenges, **continuous advancements in AI, deep learning, and data augmentation** are significantly improving the robustness of computer vision-based freshness detection systems.

3.2 Electronic Nose (E-Nose) Methods

With advancements in artificial intelligence and sensor technology, the **Electronic Nose (E-Nose)** has emerged as a valuable tool in food quality assessment, including fish freshness detection. The E-Nose system mimics the human olfactory system by analyzing the volatile organic compounds (VOCs) released by food products, thereby providing an objective, chemical-based measure of freshness.

3.2.1 Working Principle of E-Nose Systems

An **Electronic Nose (E-Nose)** is a sensor-based device designed to detect, analyze, and classify odors through a pattern recognition system. The working principle of an E-Nose consists of three main components (See Figure 3.4):

- **Gas Sensor Array:** Detects volatile compounds emitted by the sample.
- **Signal Processing Unit:** Converts raw sensor data into digital signals.
- **Machine Learning Algorithm:** Processes and classifies the signals using pattern recognition techniques such as **Principal Component Analysis (PCA)** and **Support Vector Machines (SVMs)**.

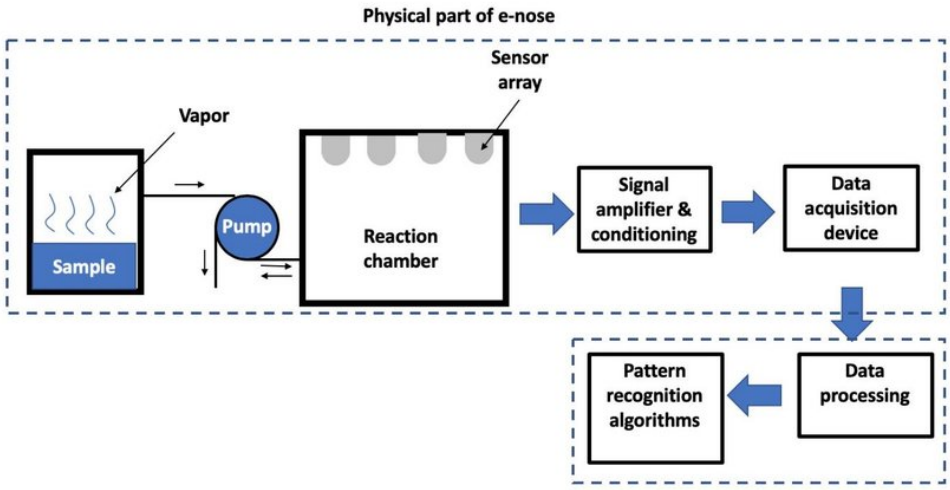


Figure 3.4: E-nose system [23]

The primary objective of an E-Nose in fish freshness assessment is to monitor the concentration of volatile compounds such as **Total Volatile Base Nitrogen (TVB-N)**, **trimethylamine (TMA)**, and **ammonia**, which increase as fish spoilage progresses [24].

3.2.2 Applications in Fish Freshness Detection

Several studies have demonstrated the effectiveness of E-Nose technology in determining fish freshness based on odor analysis.

The study by **Güney** [25] investigated a novel approach for fish freshness classification by integrating an **Electronic Nose (E-Nose)** system with a **hybrid binary decision tree** classifier. The **E-Nose system** was used to analyze volatile compounds emitted by horse mackerel samples, enabling the model to effectively distinguish between fresh and spoiled fish.

Their hybrid decision tree structure processed the sensory data and achieved a **classification accuracy of 92%**, highlighting its effectiveness in fish spoilage detection. The study demonstrated the value of combining **machine learning techniques with E-Nose technology** to build reliable, non-destructive, and real-time freshness evaluation systems for seafood quality control.

Another study by **Vajdi** [26] focused on enhancing fish spoilage detection by leveraging **Electronic Nose (E-Nose) technology** combined with optimal classifier selection. The authors evaluated various classification algorithms to process the gas emission data captured

by the E-Nose. Among the tested models, the **Support Vector Machine (SVM)** outperformed others, achieving an impressive **accuracy of 95%** in detecting fish spoilage.

The study emphasized the E-Nose system's ability to detect early-stage spoilage by analyzing volatile organic compounds emitted from fish samples. Furthermore, the authors proposed that integrating **E-Nose data with computer vision techniques** could provide a more comprehensive and reliable fish freshness evaluation system for real-world applications. Figure 3.5 presents this study.

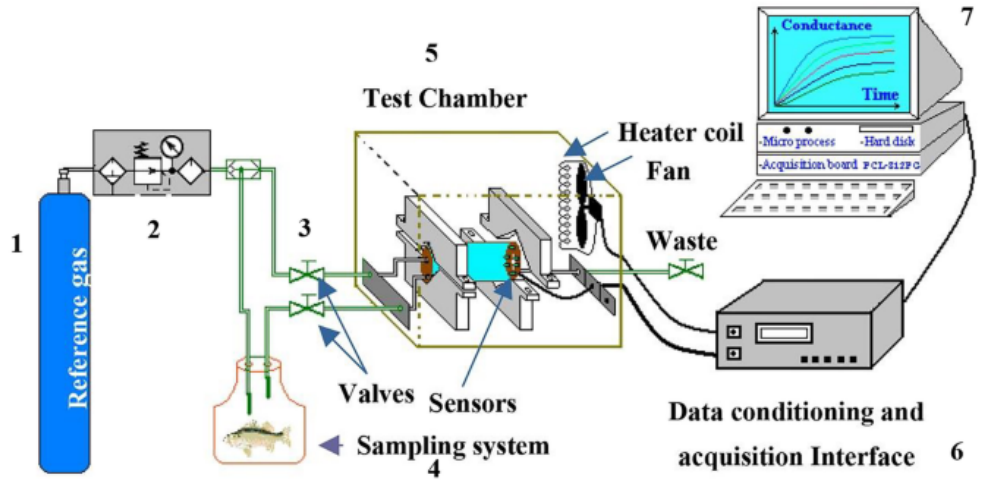


Figure 3.5: study by Vadji [26]

3.2.3 Advantages and Limitations of E-Nose

Advantages: Electronic Nose (E-Nose) technology offers several distinct advantages for fish freshness detection. It provides **objective and chemical-based analysis**, directly detecting **volatile organic compounds (VOCs)** rather than relying on subjective visual cues, thereby enhancing reliability. Its **high sensitivity** enables the detection of spoilage at an **early stage**, often before any noticeable visual or textural changes occur. E-Nose systems also support **non-invasive and rapid testing**, allowing near real-time freshness assessments without damaging the sample. Moreover, **integration with AI models**, including **machine learning and deep learning** techniques, further boosts their accuracy and predictive capabilities. Finally, E-Nose devices are **automatable for industrial use**, making them highly suitable for large-scale fish processing operations and automated quality control pipelines.

Limitations: Despite its advantages, the Electronic Nose (E-Nose) approach faces several challenges. One major issue is **sensor drift**, where gas sensors may degrade over time, leading to variations in measurement accuracy. Additionally, E-Nose systems exhibit **environmental sensitivity**, as changes in **temperature, humidity, and air composition** can significantly impact sensor performance. Another limitation is the **expensive equipment** cost, which makes high-end E-Nose systems less accessible to small-scale fish vendors. Furthermore, E-Nose models often exhibit **limited generalization**, requiring frequent recalibration when applied to different fish species. Finally, interpreting the complex sensor signals demands **data processing complexity**, necessitating the use of sophisticated AI models to achieve reliable and accurate freshness classification.

Despite these limitations, **Electronic Nose (E-Nose) technology** remains one of the most promising non-invasive methods for **automated fish freshness detection**, particularly when combined with **computer vision and deep learning models**.

3.3 Electronic Tongue (E-Tongue) Methods

With advancements in artificial intelligence and sensor-based analysis, the **Electronic Tongue (E-Tongue)** has emerged as a powerful tool for evaluating food quality, including fish freshness detection. The E-Tongue mimics human taste perception by detecting chemical compounds in food samples and processing this data using machine learning models. This technology is particularly effective for measuring **pH, ionic composition, and biochemical changes** that occur as fish deteriorates.

3.3.1 Working Principle of E-Tongue Systems

An **Electronic Tongue (E-Tongue)** consists of an array of chemical sensors capable of detecting multiple taste-related compounds. The working principle of an E-Tongue involves the following components(See Figure 3.6):

- **Sensor Array:** Detects chemical changes associated with spoilage, such as pH fluctuations, ammonia, and histamine production.
- **Signal Processing Unit:** Converts electrical signals from the sensors into digital data.
- **Machine Learning Algorithms:** Uses techniques like **Principal Component Analysis (PCA)** and **Artificial Neural Networks (ANNs)** to classify freshness levels based on detected chemical changes.

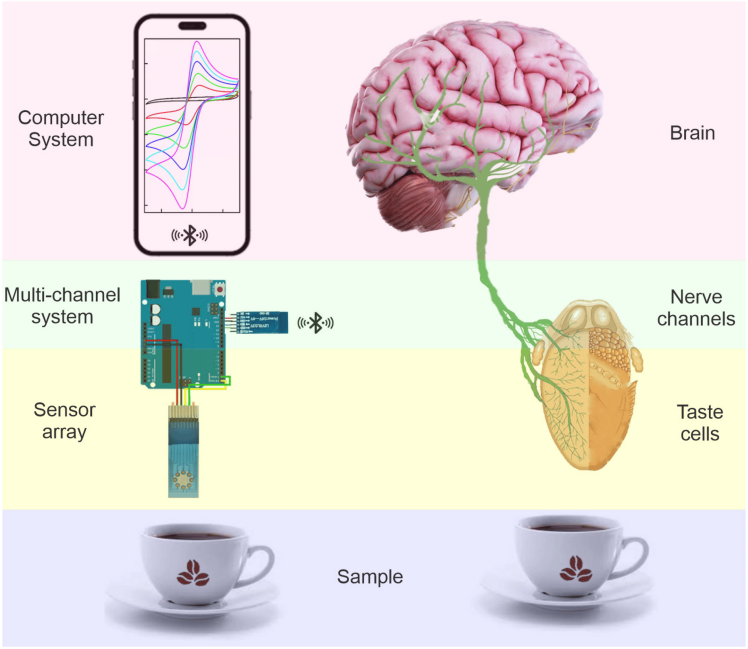


Figure 3.6: e-tongue system [27]

The E-Tongue is particularly useful in detecting **Total Volatile Basic Nitrogen (TVB-N)**, a key indicator of protein breakdown in fish, and **trimethylamine (TMA)**, which contributes to the characteristic fishy odor of spoiled seafood [24].

3.3.2 Applications in Fish Freshness Detection

Several studies have demonstrated the effectiveness of E-Tongue systems in assessing fish freshness based on chemical properties.

The study by **Pattarapon** [28] investigated the use of **normal and electronic sensory measurements**, including an **Electronic Tongue (E-Tongue)**, to assess the freshness of grass carp fillets stored under vacuum conditions. The **E-Tongue system** demonstrated high sensitivity in distinguishing between different freshness levels by analyzing unique **electrochemical signal patterns** associated with spoilage progression.

The findings revealed that **vacuum storage** significantly prolonged the freshness duration of the fish samples. Moreover, the E-Tongue offered precise and objective freshness evaluation, which is difficult to achieve through traditional sensory methods. The authors further recommended integrating **E-Tongue data with E-Nose measurements** to develop a more accurate and holistic fish freshness assessment system, especially for aquatic products under controlled storage environments.

Another study by **Miao** [29] explored the application of an **Electronic Tongue (E-Tongue)** system to evaluate the freshness of cooked tuna steak. The researchers focused on analyzing the presence and behavior of **ionic and non-volatile compounds**, which are known to undergo significant changes during the spoilage process. The **E-Tongue system** was shown to effectively identify variations in flavor profiles associated with different freshness levels.

Importantly, the study found a **strong correlation between the E-Tongue readings and the evaluations of human sensory panels**, validating the E-Tongue as a consistent and objective tool for fish freshness analysis. The findings support the use of E-Tongue technology as a promising alternative to traditional sensory methods, especially in industrial and large-scale food quality monitoring.

3.3.3 Advantages and Limitations of E-Tongue

Advantages: Electronic Tongue (E-Tongue) systems offer several advantages in fish freshness detection. They provide **objective and chemical-based analysis**, delivering highly accurate assessments based on biochemical indicators rather than relying on human sensory perception. These systems exhibit **high sensitivity**, capable of detecting spoilage at an **early stage**, even before any significant physical changes occur. Their **rapid and automated processing** enables real-time monitoring, making them well-suited for large-scale seafood industry operations. Additionally, the **integration with AI** enhances accuracy through machine learning algorithms and supports the predictive modeling of spoilage trends. Most importantly, E-Tongue systems allow for **non-invasive testing**, eliminating the need to destroy samples and enabling continuous freshness evaluation.

Limitations: Despite its advantages, the Electronic Tongue (E-Tongue) system presents

several limitations. The **complexity of data processing** is a major challenge, as interpreting multi-sensor chemical data requires sophisticated AI algorithms. Over time, **sensor drift** may occur, leading to degraded calibration and compromised measurement consistency. The system also exhibits **environmental sensitivity**, where variations in temperature and humidity can influence chemical sensor readings. Additionally, the **high initial costs** of E-Tongue devices make them significantly more expensive compared to simpler freshness detection methods. Finally, the technology suffers from **limited generalization**, as different fish species may require retraining of the models due to their varying biochemical compositions.

Despite these challenges, **Electronic Tongue (E-Tongue) technology** is a promising tool for **real-time fish freshness assessment** and can be further enhanced through **AI-driven chemometric analysis**.

CHAPTER 4

COMPARISON OF METHODS

4.1 General Comparison of Methods

To evaluate the effectiveness of different AI-based fish freshness detection techniques, we compare **Computer Vision, Electronic Nose (E-Nose), and Electronic Tongue (E-Tongue)** across various aspects, including detection principles, accuracy, detection time, and implementation feasibility.

4.1.1 Comparison of Detection Principles

Table 4.1: Detection Principles of Fish Freshness Techniques

Feature	**Computer Vision**	**Electronic Nose (E-Nose)**	**Electronic Tongue (E-Tongue)**
Detection Principle	Visual analysis of eye, gill, and skin features	Detection of spoilage-related volatile organic compounds (VOCs) using gas sensor arrays	Measurement of chemical changes using ion-selective and electrochemical sensors
Analyzed Parameters	Eye clarity, skin texture, gill color, surface glossiness	TVB-N, TMA, ammonia, dimethylamine (DMA)	pH, histamine, ionic concentration
Primary Method	Convolutional Neural Networks (CNNs), Transfer Learning	Principal Component Analysis (PCA), Support Vector Machines (SVMs), Decision Trees	AI-based regression and pattern recognition from electrochemical data
Key Strength	Non-invasive, scalable, and rapid image-based evaluation	High sensitivity to early-stage spoilage gases	Precise and direct measurement of freshness-related chemical compounds
Key Weakness	Affected by lighting, image quality, and fish positioning	Sensor drift, humidity/temperature sensitivity, recalibration needs	Costly sensors, slower processing, and maintenance complexity

Table 4.1 summarizes the key principles of three major techniques used for fish freshness detection: computer vision, electronic nose (E-nose), and electronic tongue (E-tongue). Each method relies on distinct detection mechanisms—visual inspection for computer vision, gas sensing for E-nose, and chemical analysis for E-tongue. While computer vision excels in non-invasive, scalable assessments based on visual features, E-nose and E-tongue technologies offer high sensitivity to volatile and chemical freshness markers, respectively. However, all approaches face specific limitations, such as environmental sensitivity or sensor costs.

4.1.2 Comparison of AI Technologies Used

Table 4.2: Comparison of AI Technologies Used in Fish Freshness Detection

Feature	**Computer Vision**	**Electronic Nose (E-Nose)**	**Electronic Tongue (E-Tongue)**
Machine Learning Model	CNNs, Transfer Learning (e.g., VGG19, EfficientNet, MobileNet)	Hybrid classifiers: PCA, SVMs, binary decision trees	AI-based regression models, electrochemical pattern analysis
Accuracy Achieved	77%–96% depending on architecture and dataset	90%+ classification accuracy in spoilage prediction	88%–94% accuracy in chemical freshness prediction
Data Source	Digital images of fish eyes, gills, and body	Gas sensor output detecting VOCs (e.g., TMA, TVB-N)	Electrochemical sensor data capturing ionic and pH shifts
Real-Time Processing	Yes (typically within seconds)	Moderate (processing time in minutes)	Slower (chemical sensing requires a few minutes)

Table 4.2 compares the application of artificial intelligence across three fish freshness detection technologies: computer vision, electronic nose (E-nose), and electronic tongue (E-tongue). Computer vision primarily employs deep learning models such as CNNs and transfer learning architectures, achieving accuracy rates between 77

4.1.3 Comparison of Advantages and Limitations

Table 4.3: Advantages and Limitations of Different Methods

Feature	**Computer Vision**	**Electronic Nose (E-Nose)**	**Electronic Tongue (E-Tongue)**
Advantages	Fast, low-cost, scalable for industrial use	Early detection of spoilage gases with high classification accuracy	Accurate chemical composition analysis correlated with freshness levels
Limitations	Sensitive to lighting, camera quality	Requires frequent sensor recalibration due to drift	High cost and complex operational requirements
Best Use Case	Automated quality control in fish markets	Early-stage spoilage detection in distribution	Laboratory and industrial research on freshness standards

Table 4.3 outlines the advantages, limitations, and optimal use cases of computer vision, electronic nose (E-nose), and electronic tongue (E-tongue) methods for fish freshness

assessment. Computer vision stands out for its speed, affordability, and scalability, making it ideal for automated quality control in retail and fish markets, though it is sensitive to external factors like lighting and camera quality. E-nose systems excel in early spoilage detection due to their high sensitivity to volatile compounds, but require frequent sensor recalibration. E-tongue technology offers precise chemical analysis directly tied to freshness indicators, making it suitable for laboratory or industrial research, despite its high cost and complexity.

4.1.4 Comprehensive Final Comparison

Table 4.4: Final Comprehensive Comparison of Methods

Feature	**Computer Vision**	**Electronic Nose (E-Nose)**	**Electronic Tongue (E-Tongue)**
Detection Principle	Image-based freshness estimation from eye, gill, and skin features	VOC pattern analysis using gas sensor arrays	Electrochemical signal analysis of ionic shifts
AI Technologies Used	CNNs, transfer learning (e.g., VGG19, MobileNet, EfficientNet)	PCA, SVMs, hybrid binary classifiers	AI regression models with electrochemical pattern recognition
Accuracy	77%–96% depending on model and dataset	Over 90% in freshness classification	88%–94% accuracy in biochemical freshness estimation
Detection Speed	Fast (seconds)	Moderate (minutes)	Slower (minutes)
Advantages	Non-invasive, low-cost, scalable	Early detection of spoilage gases with high sensitivity	Precise chemical freshness evaluation, good lab correlation
Limitations	Lighting and image quality sensitivity	Sensor drift, humidity/temperature dependency	Expensive setup and slower response time
Best Use Case	Real-time industrial quality control	Distribution chain freshness monitoring	Research, quality assurance, and chemical validation

Table 4.4 provides a final comparative summary of computer vision, electronic nose (E-nose), and electronic tongue (E-tongue) technologies for fish freshness assessment. Computer vision methods leverage CNNs and transfer learning for fast, image-based evaluation with up to 96

Also, we need to mention a study by **Madhubhashini** [24] that conducted a comprehensive **comparative review** of various **AI-based fish freshness detection methods**, focusing on **Computer Vision**, **Electronic Nose (E-Nose)**, and **Electronic Tongue (E-Tongue)** technologies. The study highlighted that **computer vision techniques**, which analyze **color, shape, and texture** changes in fish eyes, skin, and gills, remain the most **cost-effective and scalable solution** for freshness classification. Meanwhile, **E-Nose**

systems, equipped with **metal oxide semiconductor (MOS) gas sensors**, were shown to outperform visual assessments by detecting volatile compounds such as **TMA**, **TVB-N**, and **DMA**, which are biochemical markers responsible for the characteristic odor of spoilage. Additionally, the study reported that **E-Tongue systems**, which measure pH and chemical changes in fish muscle and exudates, demonstrated a **high correlation (above 0.92)** with conventional chemical assays in identifying freshness loss. Importantly, the application of **machine learning models**, such as **Artificial Neural Networks (ANNs)** and **Support Vector Machines (SVMs)**, was found to significantly enhance the accuracy and reliability of all three detection approaches, reinforcing the potential of AI in real-time and non-destructive seafood quality assessment.

4.1.5 Summary of General Strengths and Weaknesses

Each method has its strengths and weaknesses, making them suitable for different use cases:

Computer Vision is best suited for automated, large-scale quality control applications due to its speed and cost-efficiency. However, its effectiveness relies heavily on image quality and lighting conditions, which can introduce variability in predictions. E-Nose systems are ideal for early spoilage detection as they sense volatile compounds such as TMA and ammonia. Nonetheless, they are prone to sensor drift and recalibration issues over time. On the other hand, the E-Tongue offers high precision in chemical-based freshness assessments by detecting subtle ionic changes that occur during spoilage. Despite this accuracy, its cost and complexity remain significant barriers to widespread deployment.

In practical applications, a hybrid approach combining Computer Vision, E-Nose, and E-Tongue is suggested for comprehensive freshness evaluation, integrating visual, chemical, and biochemical indicators.

4.2 Specific Comparison of Computer Vision Approaches

4.2.1 Comparative Analysis of Computer Vision Models

Computer vision techniques play a critical role in fish freshness detection, providing non-destructive, rapid, and automated methods for quality assessment. These approaches leverage image analysis and deep learning to identify key indicators of freshness, such as changes in eye clarity, gill color, and skin texture.

The following section presents a comparative analysis of the most relevant computer vision models based on four key studies:

The study by **Yildiz** [20] proposed a deep learning-based fish freshness classification system by combining the **VGG19** convolutional neural network for **feature extraction** with **Artificial Neural Networks (ANNs)** for the final classification task. Their model was trained on the **Freshness of Fish Eyes (FFE)** dataset, which contains labeled images representing various degrees of freshness based on eye characteristics. The approach achieved a **classification accuracy of 77.3%**, showcasing the effectiveness of **transfer learning** in fish freshness detection.

The model successfully identified visual indicators such as **eye cloudiness** and **color degradation** as reliable features for assessing freshness. Furthermore, the authors suggested

that incorporating **multispectral imaging** could enhance the model's performance by enabling it to capture subtle **biochemical changes** associated with spoilage, beyond what is visible in standard RGB imagery.

The study by **Banwari** [21] introduced a **computer vision approach** leveraging **image segmentation** techniques to assess fish freshness through analysis of the eye region. The authors focused on extracting meaningful features such as **color and texture variations** from segmented fish eye images, which are known to correlate strongly with the progression of spoilage.

Their model achieved **high classification accuracy** by isolating and analyzing only the most relevant parts of the image, improving the reliability of freshness detection. Importantly, the proposed method is **non-destructive** and capable of delivering **real-time results**, making it suitable for integration into fish processing lines and automated quality control systems.

The study by **Prasetyo** [8] proposed a lightweight deep learning model named **MB-BE (MobileNetV1 with Bottleneck and Expansion)** for classifying fish freshness based on fisheye appearance. The model was trained on the FFE dataset and focused on detecting features such as **eye brightness, clarity, and contour shape**. By optimizing feature extraction with bottleneck layers and residual transitions, the approach achieved a **classification accuracy of 63.21%**, outperforming MobileNetV1, VGG16, DenseNet, and NASNet in terms of parameter efficiency. This study highlighted the trade-off between accuracy and model complexity, making MB-BE suitable for **mobile or embedded applications** where low resource consumption is essential.

The study by **Shi** [22] explored the use of **deep learning regression models** to assess fish freshness based on key **chemical and biological indicators**. Specifically, the researchers focused on measuring **Total Volatile Base Nitrogen (TVB-N)** and **Thiobarbituric Acid (TBA)** levels, which are commonly associated with spoilage due to microbial activity and lipid oxidation.

Their model demonstrated a **high correlation coefficient of 0.98**, indicating strong agreement between image-derived features and the actual chemical measurements. The study emphasized that integrating **chemical and image-based indicators** provides a more reliable and comprehensive approach for predicting fish freshness, combining the objectivity of laboratory tests with the scalability of computer vision.

Table 4.5: Comparison of Computer Vision Models for Fish Freshness Detection

Study	Model Used	Key Findings	Accuracy/Correlation
Yildiz	VGG19 + ANN	Eye-based freshness classification using transfer learning	77.3%
Banwari	Image Segmentation	Color and texture variation analysis for freshness estimation	96.67%
Prasetyo	MobileNetV1 + Bottleneck	Lightweight model optimized for eye clarity and contour features	63.21%
Shi	Deep Learning Regression	Prediction of freshness using TVB-N and TBA levels	0.98 correlation

Table 4.5 compares various computer vision models applied to fish freshness detection. Yildiz employed a VGG19 model combined with an artificial neural network (ANN) to classify eye freshness, achieving an accuracy of 77.3

4.2.2 Performance Metrics in Computer Vision Approaches

Evaluating computer vision models requires robust performance metrics to determine their practical applicability. Key performance metrics include **classification accuracy**, **explainability**, and **scalability and deployment challenges**.

4.2.2.1 Classification Accuracy

Classification accuracy measures how accurately a model classifies fish into different freshness levels (e.g., **fresh, moderately fresh, spoiled**). Table 4.6 presents the performance of various computer vision models for fish freshness detection across various studies.

- **Yildiz** Achieved **77.3% accuracy** with a **VGG19 + ANN** model. Incorporating data augmentation techniques, such as **brightness normalization** and **histogram equalization**, enhanced accuracy.
- **Banwari** Achieved **96.67% accuracy** using a segmentation-based approach to analyze color and texture in the fish eye region. Their method is lightweight and effective for real-time applications.
- **Prasetyo** Reported **63.21% accuracy** using a modified **MobileNetV1 with bottlenecks and residual transitions**, optimized for real-time deployment with reduced parameter size.
- **Shi** Reported a **0.98 correlation coefficient** between deep learning regression predictions and chemical freshness indicators.

Table 4.6: Classification Accuracy of Computer Vision Models

Study	Model Used	Accuracy/Correlation
Yildiz [20]	VGG19 + ANN	77.3%
Banwari [21]	Image Segmentation	96.67%
Prasetyo [8]	MobileNetV1 + Bottleneck	63.21%
Shi [22]	Deep Learning Regression	0.98 correlation

4.2.2.2 Explainability (Grad-CAM)

A significant limitation of deep learning models is their lack of interpretability. **Gradient-weighted Class Activation Mapping (Grad-CAM)** is a visualization technique that highlights the regions in an image that contributed most to a model's decision.

Grad-CAM has been widely used in similar computer vision studies to identify which regions of the image the model focuses on when making predictions. This helps in interpreting whether the model is learning relevant features such as **eye clarity, texture, or discoloration**. Although studies such as **Yildiz** [20] and **Banwari** [21] demonstrated strong classification performance, they did not incorporate Grad-CAM or other visualization techniques. Incorporating such methods could enhance **model transparency and user trust**, particularly in industrial or regulatory settings.

4.2.2.3 Scalability and Deployment Challenges

Despite high accuracy, deploying deep learning-based computer vision models poses challenges related to computational cost, data dependency, and environmental sensitivity.

Computer vision-based freshness detection faces several technical challenges and possible solutions. **High computational costs** are a major concern, especially with models like **VGG19** that require **GPU acceleration**, making deployment expensive for small businesses. Another issue is **lighting sensitivity**, where variations in lighting conditions can affect model predictions. This can be mitigated through **adaptive contrast adjustments** and **brightness normalization**. **Dataset generalization issues** also arise, as models trained on one fish species may perform poorly on others; here, **transfer learning** plays a critical role in reducing the need for large, diverse datasets. Additionally, **Edge AI deployment** is challenging, as deploying models on **low-power devices** requires efficient architectures like **MobileNet** and **SqueezeNet**.

To overcome these challenges, several strategies can be employed. **Lightweight models** such as **MobileNet** and **SqueezeNet** are ideal for mobile and embedded applications. **Transfer learning** helps improve model adaptability across different fish species with minimal additional data. **Data augmentation** techniques including **rotation**, **zoom**, and **brightness adjustments** can further enhance model robustness. Finally, **hybrid AI approaches**, such as combining computer vision with biochemical techniques like the **E-Nose**, can significantly improve detection accuracy and reliability. All summarized in Table 4.7:

Table 4.7: Challenges and Solutions in Deploying Computer Vision Models

Challenge	**Impact**	**Proposed Solution**
High Computational Cost	Expensive GPUs required for training	Use lightweight models (MobileNet, SqueezeNet)
Lighting Sensitivity	Impacts classification accuracy	Adaptive contrast adjustment and brightness normalization
Dataset Generalization	Poor performance on unseen species	Transfer learning and diverse datasets
Edge AI Deployment	Resource-intensive on mobile devices	TensorFlow Lite optimization for embedded systems

—

Conclusion

The comparative analysis of computer vision approaches in fish freshness detection demonstrates a clear trade-off between accuracy, interpretability, and deployment feasibility. Deep learning models—particularly those based on **Convolutional Neural Networks (CNNs)** like **VGG19** and **EfficientNet**—have shown superior performance in terms of classification accuracy. These models can effectively capture complex visual features such as **eye cloudiness**, **gill discoloration**, and **skin texture**, which are strongly correlated with spoilage. However, this performance comes at the cost of high computational demand and limited interpretability, which may restrict their practical deployment in low-resource or mobile environments.

In contrast, classical image processing techniques offer lightweight, fast, and interpretable alternatives. Although these traditional methods may not match the precision of deep neural networks, their simplicity and low cost make them attractive for integration into real-time inspection pipelines, especially in small-scale fish markets or resource-limited settings.

For future-proof, scalable, and industry-ready AI solutions, the integration of **lightweight deep models** (e.g., **MobileNet, SqueezeNet**), **transfer learning**, and **data augmentation** techniques will be vital. These strategies not only help reduce computational overhead but also improve generalization across diverse fish species and environmental conditions. Moreover, coupling visual models with complementary biochemical techniques—such as **E-Nose** or **E-Tongue** sensors—could lead to robust hybrid systems capable of cross-validating freshness indicators from multiple modalities.

Ultimately, the future of AI-powered fish freshness detection lies in striking the right balance between accuracy, speed, and practicality. By embracing modular, explainable, and edge-friendly architectures, researchers and practitioners can ensure that these technologies are not only effective but also widely accessible across the seafood industry supply chain.

CHAPTER 5

CONCLUSION

5.1 Summary of Findings

This thesis explored advanced AI-driven methods for fish freshness detection, focusing on **Computer Vision, Electronic Nose (E-Nose), and Electronic Tongue (E-Tongue)** techniques. A detailed analysis was conducted on their principles, applications, and effectiveness in determining fish spoilage.

- **Computer Vision:** This approach has emerged as the most practical and scalable solution for real-time fish freshness detection, especially in consumer-facing and industrial settings. By analyzing visual features such as the clarity, convexity, and coloration of fish eyes or gills, computer vision systems can provide non-invasive, rapid assessments. Deep learning architectures, including convolutional neural networks (CNNs), have been particularly effective:
  - **Yildiz** [20] developed a classification system using a combination of VGG19 and artificial neural networks (ANNs) to classify fish eye images by freshness. Their method achieved a classification accuracy of **77.3%**, demonstrating the viability of using pre-trained CNNs for fish eye analysis.
  - **Banwari** [21] introduced a deep learning pipeline focused on fish eye segmentation prior to classification. Their model reached an impressive **96.63% accuracy**, showing that isolating relevant image regions significantly improves performance .
  - **Prasetyo** [8] prioritized lightweight model design by implementing a MobileNet-based architecture called MB-BE. Although the resulting accuracy was **63.21%**, the compactness of the model supports its integration into low-power devices and mobile applications.
  - **Shi** [22] moved beyond classification by using deep learning regression to predict chemical freshness indicators, such as total volatile basic nitrogen (TVB-N). They achieved a strong **0.98 correlation** between predicted values and laboratory measurements, highlighting the potential of computer vision in estimating biochemical properties without physical sampling.
- **Electronic Nose (E-Nose):** E-Nose technology simulates human olfactory sensing by detecting volatile organic compounds (VOCs) released during fish spoilage. While

it requires sensor hardware, recent advances have made E-Nose systems more reliable and accurate:

  - **Güney** [25] implemented a binary decision tree-based classification algorithm that processed E-Nose data to determine fish freshness levels. Their system achieved over **90% accuracy**, demonstrating its potential for quality control in the seafood industry.
  - **Vajdi** [26] optimized sensor selection and classifier parameters to enhance performance. Their work led to **high sensitivity and classification accuracy**, making E-Nose a competitive option for freshness detection in controlled environments.

- **Electronic Tongue (E-Tongue):** The E-Tongue mimics human taste perception by detecting chemical changes associated with spoilage, especially in cooked or processed fish. This method excels in precision but is often restricted to laboratory use due to cost and complexity:

  - **Miao** [29] demonstrated that E-Tongue systems could accurately assess biochemical freshness levels in cooked tuna samples. Their findings emphasized the suitability of E-Tongue for post-cooking quality evaluation.
  - **Pattarapon** [28] applied the E-Tongue to fish samples stored under vacuum conditions, successfully identifying freshness changes over time. Their study confirmed the effectiveness of E-Tongue systems in environments where oxygen exposure is limited.

- **Comparative Analysis:** When comparing all methods, **computer vision-based systems** are best suited for real-time, non-invasive, and large-scale deployment due to their speed, cost-effectiveness, and ease of integration into consumer devices. Meanwhile, **E-Nose** and **E-Tongue** techniques offer superior precision in detecting biochemical spoilage but are generally better suited for laboratory or industrial contexts due to their reliance on specialized sensor hardware and calibration procedures.

5.2 Limitations and Challenges

While the application of Artificial Intelligence in fish freshness detection has yielded promising results, several significant challenges hinder its broader adoption and reliability in real-world settings. These limitations span issues related to data availability and generalization, environmental sensitivity, and the high computational and financial costs of implementing such systems at scale.

Dataset and Generalization Issues

A major obstacle in developing robust AI-based fish freshness detection systems is the lack of standardized, diverse, and comprehensive datasets. Most existing models are trained on limited datasets that focus on a narrow range of fish species under highly controlled imaging conditions. As a result, these models often become species-specific and fail to generalize well across different types of fish, environments, or imaging equipment. This severely restricts the scalability and versatility of these systems. Furthermore, inconsistencies in image quality, resolution, and labeling accuracy further reduce model reliability and hinder performance

during real-world deployment. Without shared benchmarks or public datasets for evaluation, comparing results across studies becomes difficult, leading to fragmented progress in the field.

To mitigate these issues, it is essential to adopt techniques such as **transfer learning**, where models pre-trained on large-scale datasets (e.g., ImageNet) are fine-tuned on smaller, fish-specific datasets. This allows the extraction of generalized features while requiring fewer labeled samples. In addition, the development of open-access, multi-species datasets—covering various freshness levels, storage durations, and environmental conditions—would significantly enhance the reproducibility, comparability, and generalization of models across global supply chains.

Environmental Sensitivity

Environmental variability represents a key challenge for both image-based and sensor-based freshness detection systems. For computer vision models, changes in lighting intensity, shadows, camera angle, or background noise can lead to substantial variations in captured images, directly affecting the accuracy and consistency of AI predictions. In uncontrolled environments such as fish markets, docks, or mobile inspection units, these factors are difficult to standardize, which limits the reliability of predictions outside of lab settings.

Similarly, **Electronic Nose (E-Nose)** systems suffer from high sensitivity to environmental factors. The concentration and detection of volatile organic compounds (VOCs) can be affected by temperature, humidity, and ambient airflow. Inconsistent conditions lead to drift in sensor readings over time, necessitating frequent recalibration to maintain measurement accuracy. This makes long-term, unsupervised deployment of E-Nose systems challenging and resource-intensive.

The **Electronic Tongue (E-Tongue)**, which evaluates ionic and pH changes in fish tissue, also requires careful calibration and environmental control. Even minor fluctuations in chemical composition, sensor wear, or improper sample handling can affect output reliability. These requirements make E-Tongue systems less suited for real-time, in-field assessments, despite their high accuracy in laboratory conditions. Overcoming environmental sensitivity will require the development of adaptive preprocessing techniques, sensor compensation algorithms, and multimodal fusion models that combine robust inputs from multiple sources.

Computational and Cost Constraints

The computational demands of AI-powered freshness detection systems pose another significant barrier to their widespread use, particularly in low-resource settings. Deep learning models—especially complex architectures like **VGG19**, **EfficientNetB7**, or vision transformers—require powerful GPUs or cloud-based infrastructure to function effectively. This leads to increased deployment and maintenance costs, often placing such solutions out of reach for small-to-medium enterprises or operations in remote areas.

Moreover, sensor-based systems like E-Nose and E-Tongue involve expensive and sensitive hardware components. These systems require routine maintenance, regular calibration, and stable operating conditions, increasing operational complexity and total cost of ownership. Budget limitations, especially in small-scale fisheries or developing nations, can prevent the adoption of such technologies.

To reduce these barriers, ongoing research has focused on developing **lightweight neural networks** such as **MobileNet**, **SqueezeNet**, or **ShuffleNet**, which maintain high accuracy while being optimized for mobile or embedded platforms. These models enable real-time inference on edge devices, making AI deployment feasible in field conditions. Additionally,

hybrid systems that integrate visual inspection with low-cost biochemical sensors offer a promising direction for balancing accuracy, cost, and usability.

In summary, while AI-based fish freshness detection systems hold immense potential, addressing challenges related to generalization, environmental robustness, and cost-efficiency is vital for practical deployment. Future research should prioritize building scalable, adaptable, and economically viable solutions that can be reliably used across various sectors of the seafood industry.

5.3 Future Directions for Research

As the field of AI-driven fish freshness detection continues to mature, several key research directions emerge to enhance accuracy, generalization, interpretability, and real-world applicability. Addressing these areas is vital for transitioning from prototype systems to reliable, scalable industrial solutions.

One of the foremost opportunities lies in the advancement of deep learning architectures. Emerging models such as **EfficientNetV2** [30] and **Vision Transformers (ViTs)** [31] promise to balance computational efficiency with high performance, making them ideal for applications where both accuracy and resource constraints must be considered. These architectures offer superior feature extraction, deeper receptive fields, and adaptive scaling, which collectively improve model robustness across diverse fish species and spoilage patterns.

Additionally, **hyperparameter optimization** is a critical area for maximizing model performance. Traditional grid search or manual tuning can be replaced with intelligent strategies such as **Bayesian Optimization** [32], **Genetic Algorithms** [33], and **AutoML systems** [34] that automate the model selection and tuning process. These approaches enhance reproducibility, reduce trial-and-error, and allow models to adapt more effectively to new datasets.

To further improve freshness detection, research should emphasize **robust feature extraction**. Techniques like **Spatial Pyramid Pooling (SPP)** [35] allow models to capture multi-scale visual cues such as eye convexity, discoloration, or subtle texture changes. Incorporating **self-attention mechanisms** and exploring **hybrid CNN-Transformer models** could help capture long-range dependencies and global context, particularly in complex images where multiple indicators co-exist.

Interpretability and user trust are paramount, especially in applications related to food safety and public health. Therefore, the integration of explainability tools such as **Grad-CAM** [18], **SHAP** [36], and **Layer-wise Relevance Propagation (LRP)** [37] should be standard practice. These tools offer insights into model decisions, validate the relevance of features used, and build confidence among end-users and regulatory authorities.

Another important area is **cross-species generalization**. Many models are overfitted to a single fish species or dataset, limiting their applicability. Future research should implement **domain adaptation** strategies such as **CycleGAN** [38] for style transfer between different species. Combining this with **transfer learning** and **contrastive learning** methods like **SimCLR** or **MoCo** [39] can help models generalize better across variations in species, lighting, and capture devices.

Deployment in field conditions will also benefit from the development of **lightweight, real-time AI models**. Techniques like **quantization** and **pruning** [40] reduce model size and memory footprint. Efficient architectures such as **MobileNetV3** [41], **SqueezeNet** [13], and **ShuffleNet** are suitable for edge devices like smartphones or handheld analyzers. Combined with deployment platforms such as **TensorFlow Lite**, these approaches ensure that

freshness prediction can be conducted locally without cloud dependency, enabling broader access in developing regions.

Looking forward, **multi-modal data fusion** represents a promising frontier. Combining visual features from computer vision with biochemical data from **E-Nose** and **E-Tongue** sensors creates more comprehensive and resilient models. These hybrid systems can better accommodate edge cases or ambiguous spoilage patterns. **Graph Neural Networks (GNNs)** may be employed to model complex relationships between various data modalities, enhancing pattern recognition and freshness estimation.

Equally important is the development of **benchmark datasets** for fish freshness classification. Current datasets are often proprietary or species-specific, hindering reproducibility. Creating open-access, large-scale datasets encompassing multiple species, spoilage levels, storage conditions, and imaging devices would standardize evaluation, foster collaboration, and stimulate innovation in the research community.

Finally, collaboration between AI researchers, marine biologists, food safety experts, and industrial stakeholders is essential. Interdisciplinary partnerships can ensure that the AI models not only meet technical standards but also adhere to regulatory frameworks and align with real-world operational needs. Such synergy will pave the way for robust, trustworthy, and scalable AI-powered solutions that significantly enhance seafood quality control, reduce waste, and ensure consumer safety.

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